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Review

Application of Artificial Intelligence and Machine Learning in Vertical Farming: A Comprehensive Review

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Abstract

Vertical farming (VF) offers a smart way to grow crops in stacked layers inside controlled indoor environments. By doing so, it uses far less land and water than traditional open-field agriculture, making it a promising solution for cities with limited space and resources. In recent years, artificial intelligence (AI), machine learning (ML), and Internet of Things (IoT) technologies have begun to transform vertical farming. These tools are moving the industry away from rigid, rule-based systems toward more flexible, data-driven operations that can adapt in real time. This paper presents a systematic review of 208 peer-reviewed studies from 2015 to 2025. It explores how AI, ML, and IoT are applied across the VF ecosystem, focusing on key areas such as computer vision for disease detection, crop growth and yield prediction, smart climate control, and precision nutrient and irrigation management. This review examines the performance of different algorithms, including Convolutional Neural Networks (CNNs), Random Forest, XGBoost, and LSTMs across hydroponic, aeroponic, and aquaponic systems. The review also covers IoT setups with multi-sensor networks, edge-cloud computing, and automated control systems. Commercial farms have shown real gains in resource efficiency and shorter supply chains. However, challenges remain: high energy use (especially from LED lighting, which makes up 40–60% of costs), expensive setup, scattered datasets, and limited real-world testing. Many high-accuracy claims (>95%) come from lab conditions and need better validation in actual farms. Overall, AI-powered vertical farming has strong potential to support resilient urban food systems. Future work should focus on lightweight edge AI models, improved data standards, explainable AI, and robust life cycle assessments to ensure the benefits outweigh the environmental and economic costs.

Keywords: vertical farming; artificial intelligence; Internet of Things; machine learning; controlled environment agriculture; precision agriculture; smart farming; crop modeling

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1. Introduction

Global food production systems face a convergence of pressures that seriously threaten the long-term sustainability of agriculture. Soil fertility is declining owing to the combined effects of global warming, indiscriminate use of agrochemicals, recurrent natural disasters, and rapid urbanization [1–3]. Together, these factors have led to widespread soil degradation and a steady reduction in per capita arable land. Climate change is accelerating these problems, further undermining the viability of agricultural systems worldwide [4,5]. Anthropogenic carbon emissions, primarily from fossil fuel combustion, deforestation,

and desertification, remain the main drivers of environmental degradation [6,7]. Simultaneously, water scarcity has become a critical bottleneck. Falling groundwater levels, inefficient irrigation practices, and widespread contamination are increasing pressure on watersheds and agricultural freshwater supplies [7]. These environmental challenges are unfolding alongside the rapid growth of the population. Projections suggest that the global population will reach 9.7–9.8 billion by 2050, requiring a sustainable increase in agricultural output of 50–60% to meet nutritional needs [8,9]. However, this expansion is severely limited by the ongoing loss of arable land. According to the Food and Agriculture Organization (FAO), per capita arable land is expected to fall below 0.20 ha by 2050, which is less than one-third of the 1970 level [8,10]. Alarmingly, arable land is disappearing at a rate of approximately 23 hectares per minute, highlighting the urgent need for better land and resource management to secure future food supplies.

Meanwhile, food production has become increasingly detached from urban consumers because of the demands of long-distance transport and distribution networks. This global food transportation system accounts for approximately 29% of total food-related greenhouse gas emissions [11]. Urban resilience, the ability of infrastructure systems to anticipate, absorb, and recover from disruptions, varies widely across regions. North America and Europe generally show the highest levels, whereas many African nations remain particularly vulnerable [12,13]. Emerging environmental stresses continue to weaken this resilience, which involves a system's capacity to mitigate, adapt to or recover from major disturbances [14–17].

Vertical farming (VF) presents a compelling alternative to address these intertwined environmental and logistical challenges. By substantially reducing land and water use, shortening supply chains, and lowering carbon emissions, VF support the broader goals of ecological preservation [18–21]. At this pivotal moment, combining sustainable VF practices with advanced technologies, such as artificial intelligence (AI), offers genuine transformative potential [22]. Exploring the synergies between AI and VF could not only boost productivity but also help shift agriculture toward more ecologically sound and resource-efficient models [23,24]. This shift also holds promise for improving food safety and public health through better urban agricultural practices.

Urban agriculture encompasses various approaches, including greenhouse, container, rooftop, and warehouse farming [25]. Among these, VF has attracted considerable attention because of its ability to produce high-value crops with high productivity and a relatively low ecological footprint [26]. In VF, crops are grown in multiple vertically stacked layers, often in underutilized urban spaces or on rooftops under tightly controlled environmental conditions [27]. The modern concept traces back to Othmar Ruthner's 1966 proposal for multi-story food production [28], but Dickson Despommier provided a comprehensive vision in 2009, proposing climate-controlled, multi-tiered buildings for horizontal or vertical crop growth [27,29,30]. By making efficient use of limited space and resources, VF offers a practical route to greater food security and environmental sustainability [31]. Its key benefits include year-round production of fresh produce, shorter transportation distances, reduced spoilage, and a significantly smaller carbon footprint than conventional supply chains [30].

The economic importance of VF is evident in its growing market share. Valued at approximately USD 5.1 billion in 2023, the sector is projected to reach USD 35 billion by 2032 [32]. Figure 1 shows these projections broken down by cultivation technology (hydroponics, aeroponics, and aquaponics), alongside a conceptual overview of the key components of a typical VF system, including soilless cultivation, nutrient delivery, automation, irrigation, lighting, CO_2 control, and environmental regulation. Regionally, East and Southeast Asia, along with North America (particularly China, Japan, Singapore, Republic

of Korea, USA, and Canada), currently hold a dominant position, accounting for over 33% of the global market [31]. Europe and the Middle East have emerged as important players in recent years [32].

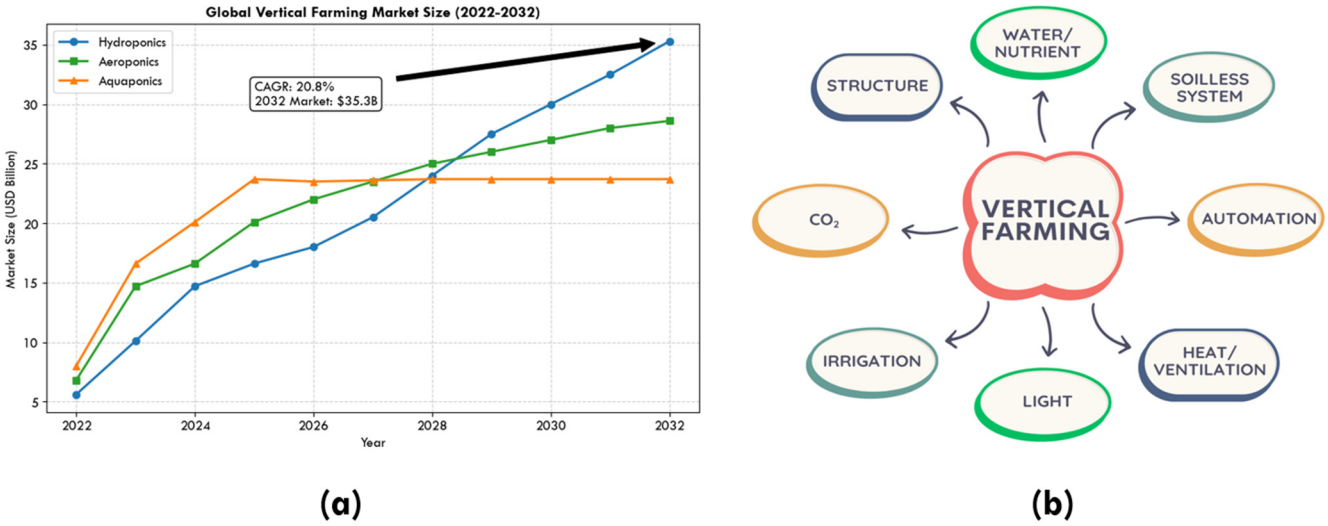


Figure 1. (**a**) Global VF market size projections from 2022 to 2032 segmented by cultivation technology (hydroponics, aeroponics, and aquaponics), indicating strong growth trends and technology adoption patterns. (**b**) Conceptual framework of a VF system showing key operational components, including structure, soilless cultivation, water–nutrient delivery, automation, irrigation, lighting, CO_2 control, and environmental regulation subsystems.

At the country level, Japan operates around 200 VF facilities producing crops such as lettuce, broccoli, tomatoes, and bell peppers, led by companies such as Spread Co., Ltd. (Kyoto) and Mirai Co., Ltd. (Gifu) [33]. China has more than 250 sites, with SANANBIO cultivating over 300 varieties of leafy greens, fruits, medicinal herbs, and flowers. Republic of Korea has built advanced infrastructure through firms such as N.Thing, NextOn, Farm 8, and ALGA Farmtech [12,33], including a prototype in Suwon that incorporates renewable energy for heating, cooling, and lighting [34]. In the United States, innovators such as 80 Acres Farms, AeroFarms, AmHydro, BrightFarms, Hort Americas, Plenty, and The Plant are at the forefront. Europe has witnessed accelerating adoption driven by consumer demand for fresh, local produce. The Netherlands leads with projects such as PlantLab, UrbanFarmers, and B-Four Agro, followed by initiatives such as Agricool in Paris and Infarm in Berlin [35]. Germany's Food4Future (f4f) program stands out for integrating the production of algae and insects with traditional crops [36].

VF marks a significant shift in agronomic science, enabling crops to be grown in vertical systems within controlled indoor environments, such as warehouses, repurposed containers, or dedicated facilities. This approach helps address many drawbacks of conventional agriculture, including high land and water requirements, heavy reliance on agrochemicals, and substantial CO_2 emissions [37,38]. With the intensification of urbanization and the reduction in arable land, VF offers a sustainable method for the continuous production of food in proximity to urban centers. A core feature of VF is its dependence on Controlled Environment Agriculture (CEA), which allows precise management of temperature, humidity, CO_2 levels, and light to optimize plant growth [38–43]. While this level of oversight assures superior yields and uniformity, it concurrently introduces intricacies in operational oversight and data handling. Although this control yields higher productivity and uniformity, it also creates considerable operational and data management complexity. AI and machine learning (ML) have become essential tools [41].

AI and ML are reshaping numerous fields by extracting insights from large datasets, uncovering hidden patterns, and supporting autonomous decision-making with limited human input [41,44]. In VF, these technologies can enhance environmental control, crop monitoring, maintenance, and farming logistics. AI systems can process data from sensors and imaging devices to support real-time decisions that improve crop health, reduce resource use, and boost overall efficiency [44,45]. ML algorithms are particularly useful for the early detection of diseases, nutrient deficiencies, and pests, allowing timely interventions that limit yield loss. They also aid in yield prediction, harvest scheduling, and labor automation [41,43,46]. As big data analytics and edge computing become more accessible, AI and ML solutions are becoming increasingly scalable and cost-effective for farms of different sizes.

Nevertheless, the integration of AI and ML into VF remains in its early stages and faces several barriers. These include the need for large, high-quality training datasets, significant upfront investments, and a lack of standardized data interoperability platforms. Successful adoption also requires interdisciplinary expertise in agronomy and data science, which can be challenging for many operators. The present review critically evaluates the current landscape of AI and ML applications in VF, discussing the technologies used, their strengths and limitations, and the likely directions for future intelligent agricultural systems.

1.1. Significance of the Present Review

Although the existing literature has thoroughly examined the general applications and benefits of VF [6,18,19,32,40,47–60], there remains a notable gap in the comprehensive integration of modern AI and IoT technologies in the specific context of vertical farming. Many previous reviews are somewhat fragmented: some focus on ML and IoT in the broader context of precision agriculture but pay limited attention to the unique characteristics of VF [55], while others explore smart farming technologies without closely linking them to high-density vertical systems [53,61]. Isolated case studies, such as the application of Random Forest for energy optimization in a single hydroponic setup [62], rarely address issues of scalability, system integration, or commercial realities at larger scales. This scattered research landscape calls for a more focused and integrated review that specifically connects AI, ML, and IoT to VF.

Furthermore, questions persist regarding the long-term ecological and economic sustainability of VF. Its dependence on energy-intensive infrastructure, potential land use conflicts, and the need for renewable energy integration raise valid concerns regarding its overall environmental benefits [40,59,60]. Without careful planning and efficiency improvements, VF's sustainability claims could be undermined. Sensor integration in hydroponic systems allows for the real-time monitoring and control of water levels, temperature, and humidity, whereas advanced algorithms enable responsive environmental adjustments [48,63]. Smart controls can reduce unnecessary interventions, save energy, and improve the responsiveness of the system. Some studies have reported notable savings, such as 20.4% in water and temperature management and 82.1% in lighting energy [24], but these findings are often context-specific and require broader validation across different settings.

This review seeks to bridge the knowledge gap by critically examining the integration of AI, ML, and IoT with VF. Unlike previous reviews, which generally addressed smart agriculture, the present study provides a comprehensive, VF-focused review of innovative systems designed to extend automation, enhance energy efficiency, reduce operating costs, and optimize environmentally friendly production practices.

The significant characteristics of this review paper are as follows:

- A systematic, PRISMA-guided methodology for literature selection and synthesis promotes transparency and reproducibility.
- A detailed analysis of IoT infrastructure and ML techniques tailored to VF, with attention to practical implementation, integration challenges, and real-world constraints, is presented.
- A comparative evaluation of AI/ML performance across different vertical farming setups, considering the suitability for various tasks, data conditions, and edge-computing environments.
- A critical assessment of commercial case studies, distinguishing verified results from company claims while evaluating economic viability, energy use, and operational risks.
- A structured discussion of key limitations, research gaps, and future directions is organized around four main pillars: data and interoperability, algorithmic advances, energy and life cycle assessment, and economic policy frameworks.

1.2. Outline of This Paper

The manuscript is structured thematically to guide readers from foundational ideas to advanced applications and real-world examples of the technology. Section 2 describes the review methodology. Section 3 introduces the key concepts of VF and controlled-environment agriculture. Section 4 provides a technical overview of IoT architectures and multilayer sensing, communication, and processing models in smart VF systems, highlighting the roles of sensor networks, edge devices, and cloud platforms in real-time monitoring and closed-loop control. Section 5 examines AI and ML applications in VF, covering computer vision for disease detection, crop monitoring, yield prediction, nutrient and irrigation management, and climate regulation, and comparing major algorithms and modeling approaches. Section 6 analyzes commercial case studies of AI-driven VF operations and assesses technology deployment, scale, and sustainability outcomes. Section 7 discusses the benefits and limitations of VF systems, including life cycle assessment perspectives. Section 8 outlines the major challenges and future research directions for intelligent VF. Finally, Section 9 concludes the review by evaluating technological readiness and priority areas for future work.

2. Review Methodology

2.1. Literature Searching Methodology

To ensure a robust and up-to-date foundation for this review, a comprehensive literature search was conducted across three major academic databases: Google Scholar, Scopus, and Web of Science (WoS). These platforms provide broad coverage, indirectly capturing relevant works indexed in ScienceDirect, IEEE Xplore, and SpringerLink. The search was limited to publications between January 2015 and December 2025. This timeframe was chosen to capture both recent technological advances in AI, ML, and IoT for vertical farming and foundational developments in this fast-evolving field.

The search strategy combined three thematic clusters of keywords related to vertical farming, artificial intelligence/machine learning (AI/ML), and the IoT. For vertical farming, terms included "vertical farm," "plant factory," "indoor farm," "controlled environment agriculture," "CEA," "hydroponic," "aeroponic," "aquaponic," and "soilless cultivation." The AI/ML cluster covered "artificial intelligence," "machine learning," "deep learning," "neural network," "computer vision," "convolutional neural network," "CNN," "long short-term memory," "LSTM," "random forest," "support vector machine," "SVM," "XGBoost," along with ensemble, reinforcement, transfer, few-shot, federated learning, and explainable AI (XAI). For IoT, keywords included "Internet of Things," "IoT," "sensor

network," "wireless sensor network," "WSN," "edge computing," "cloud computing," "fog computing," "cyber-physical system," "smart agriculture," and "precision agriculture."

These clusters were combined using Boolean operators (AND/OR) to generate precise and comprehensive results. An example search string used in Scopus was ("vertical farm" OR "plant factory" OR "indoor farm" OR "controlled environment agriculture") AND ("artificial intelligence" OR "machine learning" OR "deep learning" OR "neural network" OR "computer vision") AND ("Internet of Things" OR "IoT" OR "sensor network" OR "edge computing").

In addition to database searches, I performed a supplementary manual search by examining the reference lists of the included articles and relevant review papers. This backward snowballing technique helped identify relevant studies that might have been missed in the initial electronic search.

2.2. Eligibility Criteria

Studies were included if they met the following criteria.

- Peer-reviewed journal articles or full conference papers that presented the applications of AI, ML, or IoT technologies in vertical farming or closely related controlled-environment agriculture (CEA) systems.
- Publications accepted and indexed in recognized journals or conference proceedings.
- Works published in English.
- Studies providing empirical data, algorithmic development, systematic reviews, or case studies with adequate methodological details.
- There were no geographical restrictions; contributions from any region were considered.

Studies were excluded if they met the following criteria:

- Non-peer-reviewed items, such as editorials, commercial white papers, or conference abstracts.
- Focused solely on open-field agriculture, traditional greenhouses without a clear CEA/VF component, or general smart agriculture topics lacking specific relevance to vertical farming.
- Lacking sufficient methodological detail to evaluate the AI/ML/IoT implementation.
- Duplicate records across databases (removed during screening).

2.3. Study Selection and Data Extraction

The study selection process strictly followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines to ensure the transparency and reproducibility of the results. The initial search across the databases returned 847 records that were screened. After removing 214 duplicates, 633 unique records remained for the title and abstract screening. Of these, 312 were excluded for not meeting the inclusion criteria. Full-text assessment was conducted on the remaining 321 articles, resulting in the final inclusion of 208 primary studies.

The selection process is summarized as follows:

- Initial records identified: $n = 847$
- Duplicates removed: $n = 214$
- Records screened (title/abstract): $n = 633$
- Records excluded: $n = 312$
- Full-text articles assessed: $n = 321$
- Full-text articles excluded: $n = 113$
- Studies included in the final synthesis: $n = 208$

Data extraction was performed using a standardized form that captured key information, including bibliographic details, type of VF system (hydroponic, aeroponic, aquaponic, etc.), AI/ML algorithms used, IoT architectures and sensors, application domains, performance metrics, dataset characteristics, validation approaches, and deployment status. For commercial case studies, peer-reviewed sources were supplemented with publicly available company reports, third-party audits, and industry publications (2020–2025), while clearly distinguishing verified data from self-reported claims. During the preparation of this manuscript, the authors used Google Gemini (Google) for English grammar and language editing and to generate Figure 3 based on the authors' conceptual design and detailed instructions. The authors critically reviewed and edited all AI-generated outputs and take full responsibility for the content of this publication.

2.4. Research Questions

This review critically synthesizes the current body of research on AI, ML, and IoT applications in vertical farming. The aim of this study was to evaluate existing findings, highlight opportunities and challenges, and identify promising future directions. This study was guided by the following research questions (RQs):

RQ1—What is the current state and scope of AI, ML, and IoT integration in vertical farming? What are the main IoT architectures, sensor modalities, and communication protocols in use? Which AI/ML algorithms are most commonly applied, and for which specific tasks in VF systems?

RQ2—How effective are AI and ML algorithms in addressing key operational challenges in VF, such as disease detection, yield prediction, climate regulation, nutrient and irrigation management, and robotic automation? What performance metrics are reported, and how reliable are these results in real-world conditions?

RQ3—What are the main technical, economic, and operational barriers to the wider adoption of AI-enabled VF systems? How do issues related to data availability, interoperability, energy consumption, capital costs, and model generalization affect the current implementations?

RQ4—What are the likely future trajectories and research priorities of AI-enabled vertical farming? Which emerging technologies, such as lightweight edge-AI models, federated learning, explainable AI (XAI), and life cycle assessment frameworks, show the greatest potential for overcoming existing limitations and supporting scalable, sustainable deployment?

2.5. Synthesis and Analysis Methodology

The methodology for synthesizing the obtained data was thematic analysis. The results were classified into several categories according to the main application areas identified within the literature: (i) detection and monitoring of disease and plant well-being, (ii) growth and forecasting of crop productivity, (iii) intelligent control of fertilization and irrigation, (iv) computer vision and automatic crop monitoring, (v) climate control using AI technology, and (vi) robotics and automation. In each application area, algorithmic approaches, achieved performance results, characteristics of datasets, and methods of validation were analyzed. Whenever possible, quantitative measures (accuracy, RMSE, R^2) were collected to enable comparison. For commercially available technologies, a comparative framework was created that considered the technologies used, scale of application, productivity results, energy supply, investments, costs, and risks.

3. Overview of Vertical Farming and Technological Integration

3.1. Vertical Farming: Historical Evolution and Core Concepts

Vertical farming (VF) represents a fundamental shift in agricultural production, enabling crops to be grown in vertically stacked layers within fully controlled indoor environments. Although the idea of multi-level cultivation has ancient precedents, most famously the Hanging Gardens of Babylon, the technological foundations of modern VF have emerged gradually over the last century through a series of important innovations [27,30,64–66].

The journey began in earnest during the 1930s with William Gericke's pioneering hydroponic experiments, which demonstrated that plants could thrive in nutrient-rich water without soil. In the 1960s, Ruthner introduced greenhouse towers based on hydroponic principles, although high energy demands and maintenance costs prevented broader uptake [28]. The contemporary vision gained real momentum in the early 2000s, largely thanks to Columbia University professor Dickson Despommier, who popularized the concept of multistory urban farms as a practical response to growing food security concerns, shrinking arable land, and rising urban populations. Around the same time, Japanese researcher Toyoki Kozai advanced the field by developing closed plant production systems featuring stacked growing layers and precisely managed microclimates [27,29,64,65,67–70]. The historical timeline and technological evolution of VF from early conceptual roots to modern AI-enabled automated systems were illustrated in Figure 2.

Subsequent breakthroughs in energy-efficient LED lighting have been game-changing, providing tunable artificial light sources that are well-suited for indoor cultivation. The 2007 IPCC climate change assessment further heightened interest in VF as a climate-resilient production strategy, leading to early pilot projects in Japan, Korea, Singapore, and Sweden by 2008 [27,29,64,65,68,71–73]. Modern vertical farms integrate advanced technologies, such as IoT sensors, high-performance light-emitting diode (LED) arrays, automated nutrient delivery, and increasingly sophisticated AI-driven control systems. With the growing incorporation of renewable energy, the environmental and economic viability of VF continues to strengthen:

- Space Utilization: The crops are stacked vertically and grown inside buildings, skyscrapers, or even abandoned warehouses, resulting in a highly efficient use of land, especially for urban areas where land is extremely scarce [72].
- The Controlled Environment: The ability to carefully control the levels of light, temperature, and humidity, as well as nutrient levels, results in a highly optimized environment for crops, resulting in significantly higher crop production and significantly better crop quality compared to traditional farming methods [40].
- Water Conservation: Hydroponics and aeroponics require 70–95% less water compared to traditional farming, making this a highly viable solution to the increasingly scarce water supplies that are becoming a major problem for agricultural production and consumption [74].
- Minimized Use of Pesticides: The controlled environment means that crops are less likely to be affected by pests and diseases, resulting in a significant decrease in the use of harmful pesticides and a much more viable means for biological controls to take place [27].
- Year-Round Harvesting: The crops are grown regardless of the time of year, resulting in a highly viable means for ensuring that crops are available to the community at all times and strengthening the community through a highly consistent and predictable means for ensuring that crops are available for consumption [75].

- Lower Carbon Footprint: The close proximity to urban populations means that crops are not required to travel as far, resulting in a highly viable means for reducing the carbon footprint and increasing the level of sustainability for the environment [76].
- Healthy Crops: The controlled environment means that crops are able to thrive, resulting in a highly viable means for ensuring that crops are grown to a high level of nutritional value, especially as this is becoming a highly important factor for consumers and the products that are available for consumption [6,48,63].

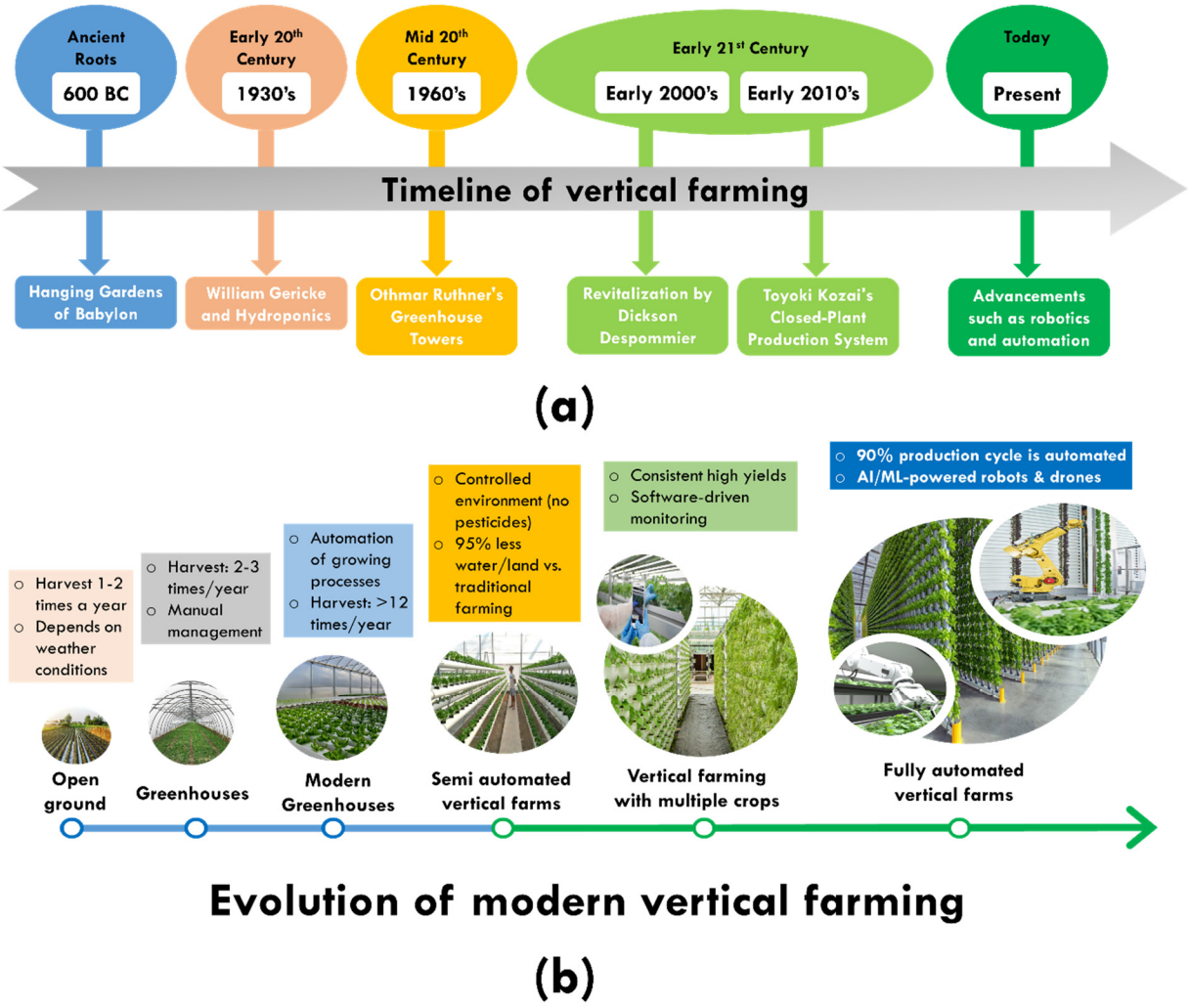


Figure 2. Historical timeline and technological evolution of VF from early conceptual roots to modern AI-enabled automated systems. (**a**) Chronological development milestones from early hydroponic concepts to closed plant production systems and robotics-enabled farms. (**b**) Transition from open-field and greenhouse cultivation to semi-automated and fully automated multi-layer vertical farms integrating software-driven monitoring and AI/ML-based control.

3.2. Types of Vertical Farming

The main cultivation approaches in vertical farming including hydroponics, aeroponics, aquaponics, and digeponics, each employ different strategies for nutrient delivery and resource cycling, offering distinct benefits and practical tradeoffs [18,19]. Figure 3 illustrates these systems side by side, highlighting the variations in nutrient delivery, root-zone design, and overall architecture. The comparative analysis of four VF nutrient delivery systems: growing medium, water usage, nutrient source, disease risk, system complexity, energy demand, and approximate market share (2022) were detailly shown in Table 1.

Table 1. Comparative analysis of four VF nutrient delivery systems: growing medium, water usage, nutrient source, disease risk, system complexity, energy demand, and approximate market share (2022).

Aspect	Hydroponics	Aeroponics	Aquaponics	Digeponics
Growing Medium	Nutrient-rich water, optional inert medium	Air with nutrient mist	Water shared between fish and plants	Digestate from anaerobic digestion
Water Usage	70% less than traditional farming	70–90% less than hydroponics	Highly efficient, recycled water	Moderate, depends on digestate processing
Nutrient Source	Synthetic or organic solutions	Synthetic solutions	Fish waste as bio-fertilizer	Digestate as bio-fertilizer
Disease Risk	Water-borne diseases possible	Minimal, no water submersion	Managed through water filtration	Low, depends on digestate quality
System Complexity	Moderate	High, precise misting is required	High, balances fish and plant needs	Moderate to high, includes waste processing
Energy Requirements	High (lighting, pumps)	Higher (high-pressure pumps)	Moderate to high (circulation, heating)	Moderate (digestion process)
Market Share (2022)	~50% (2.8B USD)	~28.6% (1.6B USD)	~21.4% (1.2B USD)	Minimal, emerging

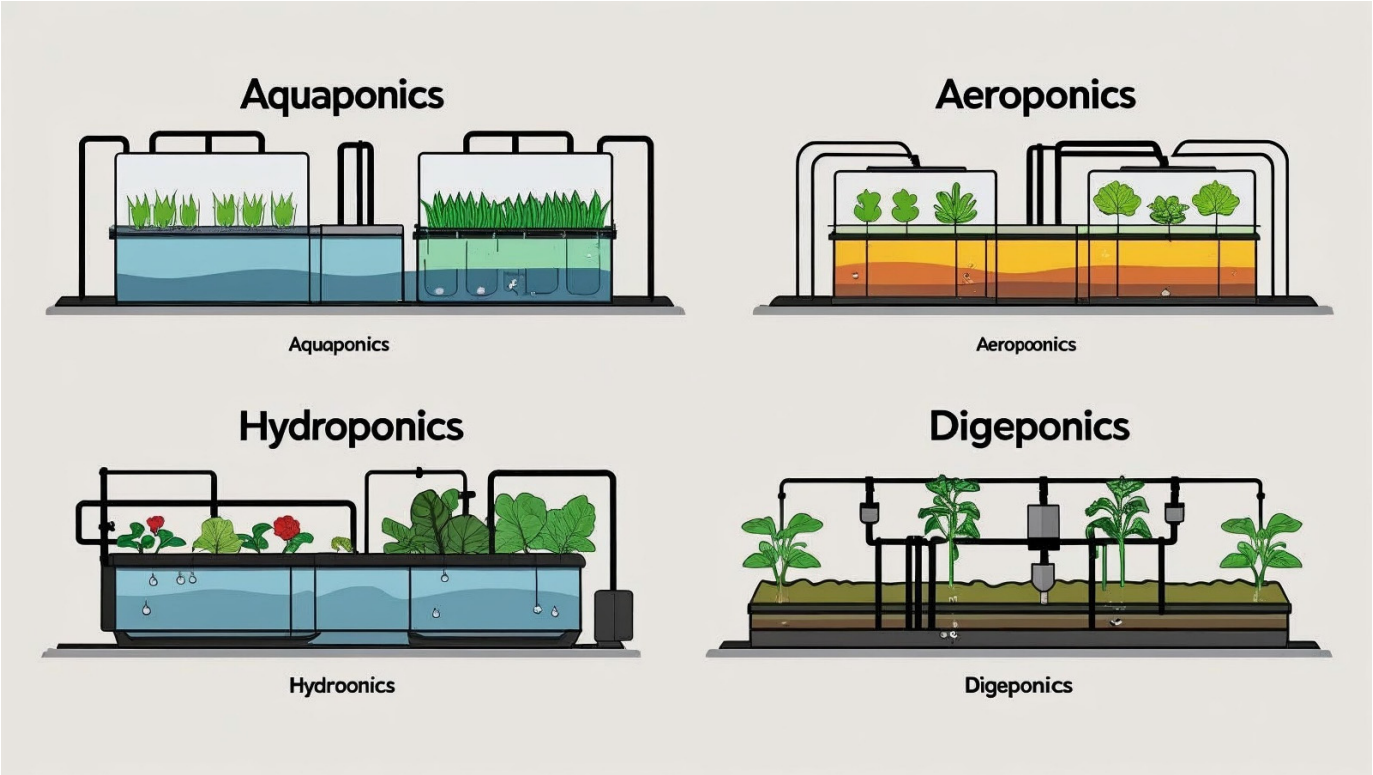


Figure 3. Comparison of major soilless cultivation techniques used in vertical farming and controlled-environment agriculture, including aquaponics, aeroponics, hydroponics, and digiponics, illustrating differences in nutrient delivery mechanisms, root-zone configuration, and system architecture.

3.2.1. Hydroponics: The Pillar of Vertical Farming

Hydroponics is the most widely used technique in vertical farming. Plants grow in nutrient-enriched water solutions, with roots supported by inert media such as perlite, rockwool, or coconut coir [77,78]. Common setups include the Nutrient Film Technique (NFT) and Deep Water Culture (DWC) This method excels with leafy greens, herbs, and strawber-

ries, often delivering yields up to 11 times higher than conventional field production for crops, such as lettuce [79–81].

Its strengths are clear: water use can be reduced by approximately 70% through recirculation, sometimes supplemented with graywater or rainwater. The soilless system minimizes soil-borne diseases and supports clean and high-quality output. These characteristics make hydroponics particularly attractive in urban environments. By 2024, it accounted for 47–65% of the VF market share, largely due to its relatively accessible setup costs and operational simplicity [6,82–84].

However, challenges remain. Upfront investment in equipment and lighting is considerable, and energy consumption can be substantially higher than that of traditional farming. Waterborne pathogens can spread rapidly without vigilant monitoring. Fortunately, advances in IoT controls and renewable energy integration are gradually alleviating these constraints [85]. Overall, hydroponics continues to serve as a reliable foundation for most commercial VF operations [79,86–90]. Advancements in renewable energy and IoT-based systems have effectively counterbalanced hydroponics' disadvantages. Hydroponics remains at the core of VF, providing ease of access and high yields [6,39,84,91].

3.2.2. Aeroponics: Water-Efficient Precision Cultivation

Aeroponics takes water use efficiency to the next level by suspending plant roots in the air and spraying them with nutrient solutions. This method was pioneered by NASA in the 1990s to grow plants in space. This method uses 90% less water than hydroponics and 95% less than traditional agriculture. Therefore, it is a very water-frugal method that is a game-changer in cities with scarce water resources [92]. The fine spray, high-pressure sprayed at 80 psi with droplets of 20 to 100 microns, is ideal for nutrient uptake and root oxygenation, leading to super quick plant growth and optimal yields [93]. Potatoes and leafy greens are among the vegetables grown using the aeroponic method, with studies indicating that water use rates are as low as 1/10 to 1/30 those of traditional field production [94,95].

The advantages of aeroponics are many, including low water consumption, which addresses global water scarcity, and the use of air, which mitigates soil- and water-borne diseases, thereby reducing pesticide use by 0–100% [96]. The use of aeroponics in VF, which falls under aeroponics, has a large market share, standing at 28.6% in 2022, making aeroponics popular in high-tech farms, as shown in Figure 1, which uses data from previous years. Despite its advantages, aeroponics is a complex and expensive cultivation method. The cost of equipment, such as high-pressure pumps and heat pipes, can be high, reaching USD 325 or more in some cases for the initial components required in aeroponics [6,39,56,97]. The energy required for aeroponics is higher than that for hydroponics, which involves only constant misting, reaching 8.46 kWh per day, making aeroponics more energy-intensive. The use of equipment in aeroponics, which requires accuracy, makes it prone to failure, where failure of a pump will result in nutrient deficiency, causing crop loss in a matter of hours if not addressed, making aeroponics less suitable for use in small-scale enterprises. Despite having some disadvantages, aeroponics remains a solution in urban agriculture, making it sustainable, especially where water needs to be conserved.

3.2.3. Aquaponics: A Symbiotic Ecosystem for Sustainable Production

Aquaponics combines aquaculture (fish farming) and hydroponics, creating a symbiotic system in which fish waste nourishes plants and cleans the water for the fish. Aquaponics is a very sustainable closed-loop system in which both crops (e.g., lettuce, herbs) and fish (e.g., tilapia) are cultivated without additional water supply [98,99]. The nutrient-rich fish waste, though high in ammonia, is used as a biofertilizer, and the crops purify the

water, reducing the need for waste disposal [97]. Aquaponics is particularly desirable for small or educational farms because it is a model of sustainable food production.

The advantage of aquaponics is its sustainability. It eschews synthetic fertilizers and relies on natural nutrient processes. Water is efficiently cycled, much like hydroponics works, and is therefore resource-efficient. The concurrent production of fish and plants differs in terms of income streams and allows farmers to adjust species to market demand [100]. Aquaponics requires fewer checks than single aquaculture or hydroponics because the system partially takes care of itself. According to Grand View Research, Aquaponics had a market share of around 21.4% in 2022, but is expected to grow at a CAGR of 20.0% due to its eco-friendly benefits. Disadvantages include substantial capital investments in fish tanks, plant beds, and water circulation systems. Power usage, particularly for water pumps and thermal control, can account for up to half of production costs [26]. Balancing the needs of fish and plants, such as water pH and oxygen levels, is complex, and system failure can harm both components [6,101]. Despite these limitations, aquaponics offers a unique model for urban farms seeking to optimize sustainability and the diversity of outputs.

3.2.4. Digeponics: Transforming Waste-to-Food Innovation

Digeponics is a recent VF method that utilizes digestate, a byproduct of anaerobic digestion, as a bio-fertilizer. Anaerobic digestion converts organic waste, such as food waste or agricultural waste, to biogas and nutrient-dense digestate, which contains essential nutrients in the form of nitrogen (0.58%), phosphorus (0.22%), and potassium (0.46%) [102,103]. Projects like Norway's "Food to Waste to Food" program demonstrate its potential by applying digestate to low-energy, bubble-insulated greenhouses with a heat loss of 0.9 W/m^2 K, compared with 5–6 W/m^2 K in conventional greenhouses [101].

The benefits of digeponics are its strategy of the circular economy. It recycles organic waste, reducing landfill use and the consumption of natural resources. The production of biogas saves energy costs and is, hence, more sustainable than other means. The process is tolerant of various crops and organic farming-friendly, which suits environmentally conscious consumers. Although there is little specific market information on digeponics because it is still in its infancy, it will rise as waste management becomes a focus area in urban agriculture. Challenges are its developmental phase, with further research necessary to optimize nutrient delivery and scale up for commercial use. Digestate quality is a function of input materials, requiring repeated processing [56,104–110]. Compounding this is the integration of the farming and digestion systems, which may incur high startup costs. Despite these drawbacks, digeponics offers a novel solution for sustainable urban agriculture, particularly in settings with waste endowments.

4. Combining IoT and AI for the Utilization of Vertical Farming

The IoT and AI technologies have become a key driving force behind the advancement of modern VF systems. Unlike traditional greenhouses or open-field agriculture, vertical farms function as highly controlled production environments where crop performance hinges on continuous monitoring, rapid feedback, and precise adjustment of environmental conditions. In this setting, IoT serves as the foundational layer for sensing, connectivity, and data collection, while AI and machine learning provide the intelligence needed to turn raw data into meaningful operational decisions [111].

Kevin Ashton introduced the term "Internet of Things" in 1999 at the MIT Auto-ID Center. This concept describes networks composed of physical items capable of sensing, identifying, and communicating with minimal human input [22,45,104,112,113]. Since its inception, the IoT has expanded swiftly across various sectors, including agriculture. Currently, billions of devices are estimated to be interconnected globally. In the context of vertical farming,

this results in extensive networks of sensors, controllers, and actuators distributed throughout the growing racks, chambers, and nutrient systems [22,45,104,112–115].

In practical VF operations, IoT and AI are seldom employed independently of each other. Rather, they integrate into a cohesive system: sensors collect data regarding environmental and plant conditions, communication networks relay these data, and edge and cloud platforms manage their storage and processing. Subsequently, the AI models examine the patterns to inform the control actions. This results in a closed-loop system in which monitoring, forecasting, and actuation function in harmony [54,108,114–118].

In engineering terms, the integration process generally consists of four primary phases: sensing multiple parameters and gathering data, communication and aggregation through networks, processing data using AI analytics, and control and visualization at the application layer. To ensure scalability and interoperability, most smart VF systems adopt a layered IoT architecture [45,105,108–110,115,118–123]. The framework of integrating IoT and AI for VF operation were shown in Figure 4.

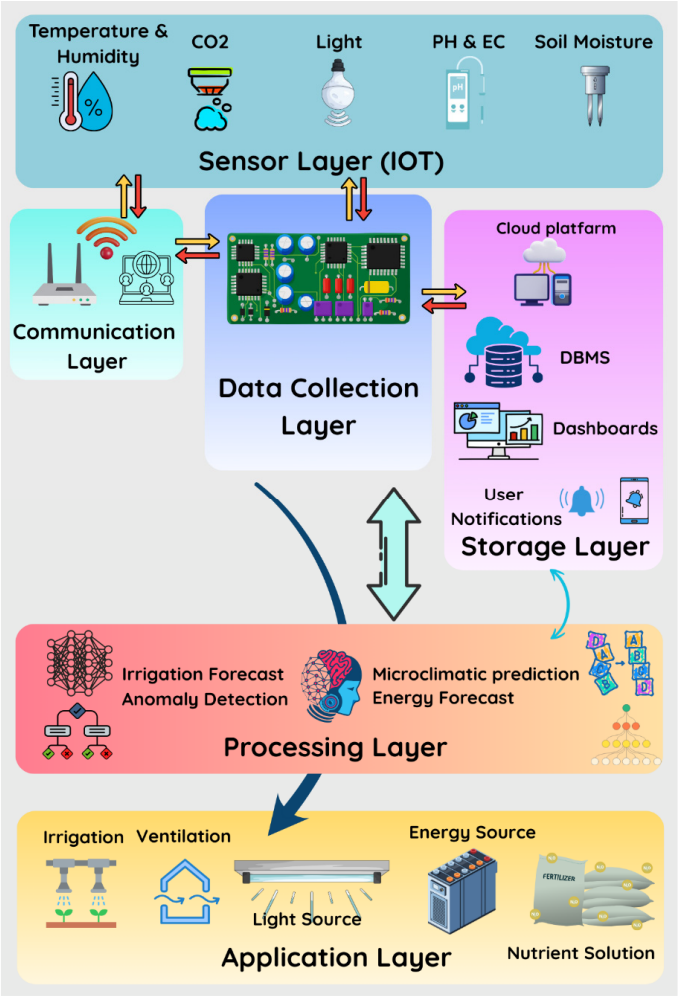


Figure 4. Layered architecture of an IoT- and AI-enabled VF system showing sensor acquisition, communication and data collection, cloud storage, AI-based processing, and automated application control. Source: Developed by the authors.

4.1. Technical IoT Infrastructure for AI-Enabled Vertical Farming

4.1.1. Sensing Layer and Instrumentation

In AI-driven vertical farming (VF), high-resolution sensing is crucial for data gathering. Sensor arrays are strategically placed at different levels, such as racks, trays, and zones, to monitor the microclimate, nutrient solutions, and air quality. The key variables monitored

include air temperature, relative humidity, vapor pressure deficit (VPD), CO_2 levels, photosynthetic photon flux density (PPFD), pH, electrical conductivity (EC), dissolved oxygen, and particulate matter [108,113,118,122,124,125].

Popular sensors for measuring temperature, humidity, and gases include Bosch BME680, Sensirion SHT31/SCD30, and DHT22, while NDIR modules such as MH-Z19C are used for CO_2, and Plantower PMS5003 is employed for particulate matter detection. Nutrient systems incorporate industrial pH and EC probes, and optical sensors, including RGB, multispectral, and thermal cameras, assist in plant phenotyping and in the identification of stress. The selection of sensors is influenced by the type of crop, growth stage, and spatial variability within the farm. Typically, signals are digitized and pre-processed at the edge before undergoing a more detailed analysis [110,113,120,124,126].

4.1.2. Edge Devices and IoT Gateways

Sensor nodes typically connect to edge controllers and gateways (such as ESP32, Raspberry Pi, or NVIDIA Jetson Nano) that aggregate, timestamp, and perform the initial processing of data streams. These devices enable local filtering, sensor fusion, and basic rule-based control [45,108,110,113,115,118,119,125,127].

Gateways act as bridges to cloud services and handle data normalization, batching, and protocol translation. Many studies also support lightweight AI inference for real-time anomaly detection and decision-making [109,110,113,124].

Common communication technologies include Wi-Fi, Ethernet, ZigBee, Bluetooth Low Energy, LoRaWAN, NB-IoT, and LTE-M. Protocols like MQTT, CoAP, and AMQP are favored for their efficiency in publish–subscribe models, which enhance scalability [45,56,104,107–110,113,115,118–120,126,127]. Publish–subscribe messaging models are preferred because they decouple sensing nodes from analytics services, thereby improving scalability.

4.1.3. Layered IoT Architecture

Most AI-integrated VF systems adopt a multilayer architecture to separate concerns and improve modularity. A five-layer model is frequently referenced in the smart agriculture literature [32,48,56–59,61,104,128–132].

Perception Layer: This layer includes physical sensors and actuators interfacing directly with the crop environment. Devices convert physical signals into electrical outputs and execute control actions such as valve switching, pump control, LED dimming, and ventilation adjustment.

Network Layer: The network layer manages data transport between edge devices, gateways, and cloud endpoints. It operates using IP-based stacks built on TCP/IP, UDP, and secure transport protocols such as TLS/SSL. In dense indoor farms, network design must address latency, packet loss, and interference from metallic rack structures and LED drivers.

Data Processing Layer: This intermediate layer performs stream ingestion, filtering, aggregation, and transformation. Edge and fog nodes execute preprocessing workflows to reduce bandwidth load and enable rapid control loops. Containerized microservices and stream processors are increasingly used to handle high-frequency telemetry.

Storage Layer: Persistent storage is typically implemented using mixed database architectures. Time-series data are frequently stored in engines such as InfluxDB or TimescaleDB, while structured operational data are managed through relational systems such as MySQL and PostgreSQL. Local buffering using SD card or eMMC storage is often configured to provide redundancy during network outages. Distributed object storage systems are used for image and video datasets required by computer vision models.

Application Layer: The application layer exposes user and control services through dashboards, analytics portals, and supervisory control interfaces. Visualization and moni-

toring are commonly implemented using platforms such as Grafana, Kibana, and ThingsBoard, or custom web dashboards built on modern frameworks. These interfaces support trend visualization, threshold alarms, rule engines, and actuator control panels. Notification services are integrated through email, SMS, and mobile push channels to support rapid operator response.

4.2. Role of AI in Enhancing IoT Capabilities

Artificial intelligence augments IoT in VF by transforming raw sensor data into predictive models and automated decision-making processes. ML algorithms can analyze historical and real-time data to optimize irrigation, nutrient delivery, lighting schedules, and even yield forecasting [32,56,133,134]. For instance:

- AI models can predict optimal harvest times based on environmental data trends.
- Anomaly detection algorithms identify system faults or disease outbreaks at early stages.
- RL and deep neural networks can learn to adapt to evolving conditions, increasing operational efficiency and reducing resource waste.
- If combined, IoT and AI systems implement closed-loop feedback loops in VF. For instance, data gathered in the perception layer by sensors is processed and translated by AI models in the application layer, which then sends feedback back to actuators for real-time system adjustment, establishing a self-optimizing agrarian environment.

4.3. Theory and Design of AI Algorithms for Vertical Farming

4.3.1. Role of Artificial Intelligence in Vertical Farming Systems

AI and ML serve as the computational core of modern VF, transforming large volumes of heterogeneous data generated by sensors and imaging systems into actionable insights for environmental control, crop monitoring, disease detection, and yield prediction [32]. Because VF operates in tightly controlled environments with high-frequency data and strong variable interdependencies, the algorithm design must carefully consider the data characteristics, control requirements, and computational limits [24,32,129,135–137]. As a result, AI-driven decision frameworks are central to improving land use efficiency, resource optimization, and operational sustainability in VF systems.

4.3.2. Machine Learning Paradigms Applied in Vertical Farming

Machine learning techniques used in VF can be broadly categorized into supervised, unsupervised, and reinforcement learning approaches, each addressing distinct operational requirements.

Supervised Learning: Supervised learning algorithms are widely applied in VF, as labeled datasets are readily available from historical yield records, sensor measurements, and annotated plant images. These methods are commonly used for regression and classification tasks, such as crop yield estimation, nutrient status prediction, and disease identification. Linear and logistic regression models provide interpretable baselines for yield and health assessment, while more advanced methods such as Support Vector Machines (SVMs), Random Forest (RF), and Extreme Gradient Boosting (XGB) offer improved predictive performance for complex, non-linear relationships. Ensemble models, particularly RF and XGB, are frequently reported in VF studies due to their robustness to noisy sensor data and ability to capture interactions among environmental variables [129,136–138].

Unsupervised Learning: Unsupervised learning techniques are employed when labeled data are limited or unavailable. In VF applications, clustering algorithms such as k-means are used to group plants or cultivation zones based on growth patterns, nutrient uptake, or microclimate conditions. Dimensionality reduction techniques, such as Principal

Component Analysis (PCA), help identify the dominant factors influencing crop growth and system performance. These approaches are particularly useful for exploratory data analysis and anomaly detection in complex VF environments [129,133,139].

Reinforcement Learning: Reinforcement learning (RL) focuses on sequential decision-making through interactions between an agent and its environment. In VF, RL has potential applications in adaptive control tasks, such as optimizing lighting schedules, irrigation timing, and climate regulation strategies. By learning policies that maximize long-term rewards, RL-based controllers can dynamically balance crop productivity and energy consumption. Although still emerging in VF research, RL represents a promising approach for developing autonomous control systems [129,136–138].

4.3.3. Deep Learning and Machine Vision in Vertical Farming

DL methods have gained prominence in VF due to their ability to process high-dimensional and unstructured data, particularly images and time-series sensor data. Convolutional Neural Networks (CNNs) are the dominant architecture for machine vision tasks, including disease detection, growth stage classification, and leaf area estimation. By automatically learning hierarchical visual features such as color, texture, and shape, CNNs reduce the need for manual feature engineering and enable accurate image-based analysis under controlled lighting conditions typical of VF systems [129,136–138,140,141].

For temporal modeling, Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are used to capture dependencies in sequential environmental data, supporting applications such as climate forecasting and predictive control. To address computational constraints in real-time VF operations, lightweight architectures such as MobileNet variants are increasingly adopted for deployment on edge devices, enabling on-site inference with reduced latency and energy consumption [22,128,133,134,142–145]. The Common ML Algorithms, along with the application, merits, and demerits for VF applications from the previous studies, are mentioned in Table 2.

Table 2. Common Machine Learning Algorithms in Vertical Farming.

Algorithm	Type	Application in VF	Advantages	Disadvantages
Linear Regression	Supervised	Yield prediction	Simple, interpretable	Assumes linear relationships
Logistic Regression	Supervised	Disease classification	Effective for binary outcomes	Limited to linear boundaries
Decision Trees	Supervised	Nutrient management	Intuitive, mimics human decision-making	Prone to overfitting
Random Forest	Supervised (Ensemble)	Yield prediction, disease detection	Handles large datasets robust	Computationally intensive
Support Vector Machines	Supervised	Disease detection	Robust to outliers	Memory intensive
Convolutional Neural Networks	Deep Learning	Image-based disease detection, growth monitoring	Automatic feature extraction, high accuracy	High computational demands
Deep Neural Networks	Deep Learning	Yield prediction, environmental control	High accuracy for complex data	Requires large datasets
Extreme Gradient Boosting	Supervised (Ensemble)	Yield prediction, nutrient control	High performance, handles missing data	Risk of overfitting
k-Means Clustering	Unsupervised	Plant growth clustering	Identifies patterns without labels	Sensitive to initial conditions
Principal Component Analysis	Unsupervised	Data dimensionality reduction	Simplifies complex datasets	Loss of interpretability

4.3.4. Data Pipeline for AI-Driven Vertical Farming

The performance of AI models in VF is strongly influenced by the quality and structure of the underlying data pipeline. Data collection typically involves environmental sensors measuring temperature, humidity, CO_2 concentration, and light intensity; nutrient sensors monitoring pH and electrical conductivity; and imaging systems capturing high-resolution plant images. These data streams are often integrated through IoT platforms to support continuous monitoring.

Preprocessing is essential to ensure data reliability and model robustness. Common steps include data cleaning to remove noise and outliers, normalization or standardization of sensor readings, and augmentation of image datasets through transformations such as rotation or scaling. Feature engineering may be applied to derive additional indicators, such as growth rates or nutrient uptake trends, from raw measurements [23,129,136–138,146,147].

Feature selection techniques play a critical role in reducing model complexity and improving interpretability. Methods based on feature importance scores from ensemble models, as well as explainable AI tools such as SHAP, are increasingly used to identify key environmental and physiological drivers of crop performance. In contrast, DL, particularly CNNs, perform implicit feature extraction, minimizing the need for manual selection while benefiting from explainability frameworks to support decision transparency [23,129,136–138,146,147].

4.3.5. Model Training, Evaluation, and Deployment Considerations

Reliable evaluation of machine learning models is essential in VF applications because AI-driven decisions directly influence irrigation control, nutrient dosing, disease detection, and yield forecasting. Model performance is therefore not assessed using a single indicator but through a set of quantitative metrics that measure prediction accuracy, error magnitude, and generalization capability on unseen data [22,128,133,134,142–145]. In VF studies, the choice of metric typically depends on whether the task is formulated as a regression problem (e.g., yield or growth prediction) or a classification problem (e.g., disease detection or stress identification).

For regression-oriented VF applications such as biomass estimation, evapotranspiration prediction, and microclimate forecasting error-based metrics are most frequently reported. Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are widely used because they directly quantify deviation between predicted and observed values in physical units. Mean Square Error (MSE) and Sum of Squared Errors (SSE) are also applied during model optimization due to their sensitivity to large deviations. In addition, the coefficient of determination (R^2) is commonly used to express goodness-of-fit, with values closer to one indicating stronger explanatory power. Several VF modeling studies also report adjusted R^2 to compensate for bias introduced by multiple input variables [23,129,136–138,146,147].

Classification-based VF tasks, including pest detection, leaf disease recognition, growth-stage classification, and anomaly detection, are evaluated using confusion-matrix-derived metrics. Accuracy provides an overall correctness measure but can be misleading in imbalanced datasets, which are common in early disease detection scenarios. Therefore, precision and recall are frequently reported alongside accuracy. Precision reflects how many predicted positive cases are truly positive, while recall (true positive rate) measures how completely the model captures actual positive events. The F1-score, defined as the harmonic mean of precision and recall, is often preferred when balancing false positives and false negatives is critical [22,128,133,134,142–145].

Receiver Operating Characteristic (ROC) analysis and the Area Under the Curve (AUC) are increasingly used in AI-based plant health and stress detection models to evaluate discrimination capability across decision thresholds. Additional indicators such as True

Negative Rate (specificity), False Positive Rate, and log-loss are reported in probabilistic classifiers and DL vision models used in VF monitoring systems [23,129,136–138,146,147].

Across recent VF literature, best practice is shifting toward multi-metric reporting, where at least one scale-dependent error metric (such as RMSE), one normalized goodness-of-fit metric (R^2), and one class-sensitive metric (such as F1-score or AUC) are presented together [22,54,148]. The performance metrics commonly used for regression models in AI-enabled vertical farming systems were mentioned in Table 3 whereas Table 4 have the performance metrics commonly used for Classification models. This combined evaluation approach provides a more robust picture of model reliability under real operational conditions.

Table 3. Performance metrics commonly used for regression models in AI-enabled vertical farming systems.

Metric	Formula	Interpretation	Typical VF Use
Error (E)	$E_i = A_i - P_i$	Difference between observed and predicted value	Base residual for model diagnostics
Mean Absolute Error (MAE)	$MAE = (1/n)\ \Sigma$	$A_i - P_i$	
Sum of Squared Errors (SSE)	$SSE = \Sigma\ (A_i - P_i)^2$	Total squared residual error	Model fitting objective
Mean Squared Error (MSE)	$MSE = (1/n)\ \Sigma\ (A_i - P_i)^2$	Average squared error	Penalizes large deviations
Root Mean Square Error (RMSE)	$RMSE = \sqrt{[(1/n)\ \Sigma\ (A_i - P_i)^2]}$	Error in original unit scale	Climate and growth forecasting
Correlation Coefficient (R)	$R = corr(A, P)$	Linear association strength	Trend agreement check
Coefficient of Determination (R^2)	$R^2 = 1 - [\Sigma\ (A_i - P_i)^2 / \Sigma\ (A_i - \bar{A})^2]$	Variance explained by model	Overall regression fit quality
Adjusted R^2	$R^2adj = 1 - (n - 1)/(n - k - 1)$	R^2 corrected for predictors	Multi-feature VF models

Symbols: A_i = actual value, P_i = predicted value, $\bar{A}$ = mean of actual values, n = samples, k = predictors.

Table 4. Performance metrics commonly used for classification models in AI-based vertical farming applications.

Metric	Formula	Interpretation	Typical VF Use
Accuracy	(TP + TN)/(TP + TN + FP + FN)	Overall correct predictions	General model comparison
Precision	TP/(TP + FP)	Correct positive prediction rate	Pest/disease alerts
Recall (TPR)	TP/(TP + FN)	Detection completeness	Early stress detection
Specificity (TNR)	TN/(TN + FP)	Correct negative detection	False alarm control
F1-Score	2TP/(2TP + FP + FN)	Balance of precision & recall	Imbalanced crop datasets
False Positive Rate	FP/(FP + TN)	False alarm proportion	Anomaly detection systems
ROC Curve	TPR vs. FPR plot	Threshold behavior	Classifier evaluation
AUC	Area under ROC	Class separability	Model ranking
Log Loss	$-\Sigma\ y\ log(p)$	Probabilistic error	DL classifier training
MCC	$(TP \cdot TN - FP \cdot FN)/\sqrt{((TP + FP)(TP + FN)(TN + FP)(TN + FN))}$	Balanced quality score	Imbalanced VF classes

Symbols: TP = true positive, TN = true negative, FP = false positive, FN = false negative.

4.4. Benefits and Real-World Applications

The combination of IoT and AI in vertical farming provides various benefits:

Resource Optimization: The automated management of lighting, irrigation, and nutrient dosing reduces the consumption of water and energy.

Predictive Maintenance: Equipment failures can be forecasted and scheduled by AI-driven analytics.

Scalability: Modular IoT facilitates simple expansion of farm infrastructure.

Remote Monitoring: Users can monitor farm conditions and respond remotely through mobile apps or web portals.

Practical applications of such systems are now common in urban farming initiatives, where sensors track the status of plants in real time, and AI platforms continuously change system parameters to optimize crop output. These technologies together improve the quality of the yield, lower labor expenses, and encourage sustainable agriculture.

The application of AI and ML in VF can be understood as a closed-loop intelligent control ecosystem where sensing, analytics, prediction, and automated actuation operate continuously. A generalized intelligent VF operational model illustrating this AI-driven sensing–decision–reaction cycle is shown in Figure 5.

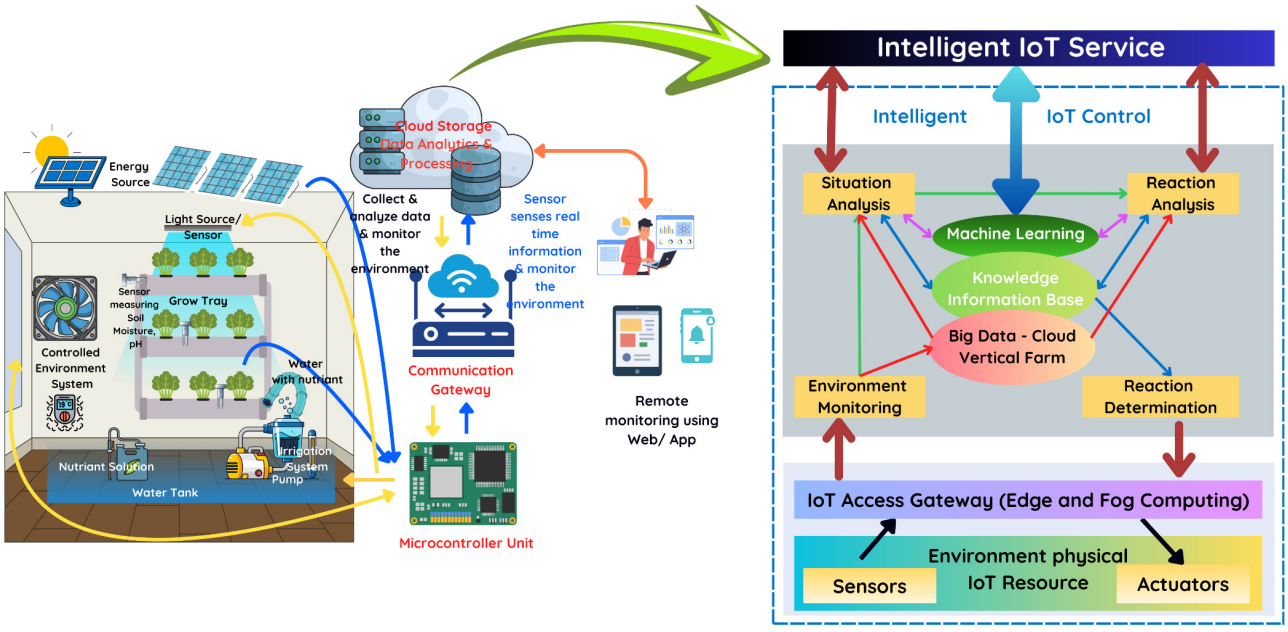


Figure 5. Integrated intelligent VF model combining renewable energy input, multi-layer crop production, sensor-based environmental monitoring, automated nutrient and irrigation control, cloud data storage, real-time analytics, and AI-driven decision support with remote monitoring via web and mobile interfaces.

5. Application of Artificial Intelligence and Machine Learning in Vertical Farming

AI and ML technologies are increasingly recognized as foundational enablers of next-generation VF systems [22,54,75,148]. Unlike conventional agricultural automation, which typically relies on threshold-based control and manual supervision, AI-driven VF operates through continuous data acquisition, predictive analytics, adaptive control, and automated decision execution. The reviewed literature demonstrates that AI and ML are not applied as isolated analytical tools but are embedded across the full operational pipeline of vertical farms, including environmental control, crop health monitoring, yield prediction, nutrient management, irrigation automation, and robotic operations [22,54,75,148,149]. VF environments generate large volumes of heterogeneous data originating from IoT sensors, imaging systems, nutrient monitoring devices, and climate control subsystems. AI and ML models transform these multi-source datasets into actionable intelligence by learning patterns, detecting anomalies, predicting crop responses, and optimizing control strategies. Across the surveyed studies, AI applications in VF can be grouped into major functional domains: disease and stress detection, crop growth and yield prediction, intelligent nutrient

and irrigation control, environmental optimization, computer vision–based monitoring, and robotic automation [150].

5.1. Disease Detection and Plant Health Diagnosis Using AI and ML

In VF systems, especially in hydroponics, aeroponics, and aquaponics, one of the most studied applications of AI is the early and accurate detection of plant diseases. VF systems have high planting density, and any undetected disease outbreak can spread quickly. Traditional manual scouting and expert diagnosis are often impractical for continuous indoor production cycles. AI-based image analytics and ML classification models play a critical role in automated plant health surveillance. Most of the presented techniques are based on camera-based image acquisition connected to IoT platforms where images of leaves and canopies are acquired and processed continuously using ML and DL algorithms. CNNs have dominated the disease detection tasks because of their strong visual feature extraction and classification performance. Several studies have reported CNN and deep CNN variants at accuracy levels often above 95% in identifying crop diseases in lettuce, tomato, apple, grape, and other VF crops. For instance, a knowledge-distillation-based DCNN model achieved 99.4% accuracy with only 6 W power consumption on edge devices in smart hydroponics [151], while a DLCNN-based smart hydroponics expert system achieved 99.82% for classification and 99.297% for prediction of nutrient-adjusted disease responses [152]. Lightweight architectures such as MobileNet variants have been successfully deployed on edge devices (Raspberry Pi) to reduce the computational load while still maintaining high detection accuracy, one MobileNet V2 system on a robotic hydroponics platform achieved 98.19% accuracy for lettuce disease detection [153,154]. Traditional CNN models are also used, apart from object detection methods like Faster R-CNN, YOLO and its variants, and DeepLab segmentation models, to detect diseased areas, abnormal leaves, and sub-health plant areas in hydroponic systems. U3-YOLOWs obtained 94.38% mAP for sub-health areas in rape, while DeepLabV3+ with ResNet-101 obtained 99.24% pixel accuracy for abnormal leaf segmentation in lettuce [155,156]. The ensemble and hybrid approaches that integrate the segmentation with classification further enhance the diagnostic precision. SVM, RF, KNN, and Naïve Bayes classifiers are also more commonly reported, particularly when handcrafted feature extraction such as color histograms, texture descriptors, and shape metrics are extracted before classification. For apple fruit disease SVM models achieved 94.1% and for healthy vs. chlorotic lettuce in rotating hydroponics 100% [146]. In some implementations disease detection models are integrated directly in edge controllers like Raspberry Pi platforms which allow near real-time classification and actuator response. IoT dashboards and mobile applications send alerts upon disease detection that help farmers take specific actions like nutrient adjustments or localized pesticide applications. The AI-based disease detection systems reduce labor requirements and allow earlier interventions compared to manual inspection, thus reducing yield loss and excessive chemical use [157].

Critical Assessment: While the accuracy numbers are striking, a few important caveats should be kept in mind. Most studies rely on controlled datasets with limited variability (typically fewer than 5000 images) captured under ideal lighting. Real-world performance tends to drop due to sensor drift, differences between cultivars, and varying light spectra from different LED combinations. Cross-validation across farms and seasons is rarely performed. Additionally, many papers do not sufficiently address class imbalance, which is especially problematic in early disease detection where healthy samples greatly outnumber diseased ones. Transfer learning and data augmentation offer promising ways forward, but systematic benchmarking under diverse operational conditions is still largely missing.

5.2. AI-Driven Crop Growth Monitoring and Yield Prediction

Another important use of AI in VF is to model growth and predict yields. VF is operated under controlled conditions, which provides high-resolution environmental and crop datasets, making the environment especially suitable for predictive modeling. ML and DL models learn relationships between environmental variables, nutrient parameters and plant growth indicators, to predict yield, biomass accumulation, and harvesting time. Random Forest, SVR, XGBoost, Decision Trees, and DNNs are typical supervised learning models for yield prediction in hydroponic and aeroponic systems. For lettuce yield in three types of hydroponics over two years, the best results were obtained using XGBoost with an R^2 of 0.94 and an RMSE of 8.88 [145]. For tomatoes in aquaponics, an ANFIS model had an R^2 of 0.9918 and an RMSE of 0.4582 [158]. The economic viability of AI-assisted nutrient control systems for small-scale operators remains uncertain. Ensemble methods and gradient boosting approaches are more robust to non-linear interactions and noisy sensor data and tend to report better predictive accuracy in the literature. For VF yield prediction, performance metrics reported in the literature tend to be high in terms of coefficients of determination ($R^2 > 0.9$) and low in terms of RMSE if sufficient training data are available. DL methods also improve yield prediction capability through the integration of image-based features. Non-destructive growth monitoring is enabled by estimating leaf area, canopy coverage, and plant size from RGB and RGB-D images using CNN and Mask R-CNN models; a two-stage ResNet50V2 CNN achieved an R^2 of 0.88–0.95 and RMSE 6.09% for lettuce growth indices [159], and Mask R-CNN reached 97.63% mean accuracy for leaf area estimation with images captured every 30 min [160,161]. Dual-stage CNN architectures and ResNet-based networks have been employed to estimate fresh weight, dry weight and morphological traits with excellent predictive performance. RNNs and LSTM models are suitable for time-series environmental data, capturing temporal dependencies in climate and nutrient dynamics for growth forecasting. AI-driven growth models also support operational planning by forecasting optimal harvest times, enabling staggered harvesting approaches and better supply stability. Many studies highlight that predictive models support policy and operational decisions by predicting production under different environmental control strategies. An ensemble DNN + RF model achieved an R^2 value of 0.81 for multiple crops in aeroponics. Random Forest achieved 99.27% accuracy for multi-crop yield prediction [62,149].

Critical Assessment: The R^2 values reported by many studies are often high because they are obtained under controlled conditions with less environmental variation. The predictive ability of these models under real-world operational variability, including sensor drift, equipment failure, and crop heterogeneity, is under-explored. Moreover, most studies are limited to single crops (mainly lettuce), and it remains to be tested whether the results are generalizable to other species. Furthermore, yield prediction is often treated too simply in terms of time scale, as long-term (weeks ahead) forecasting is still much more complex than short-term estimation. In the future, cross-crop validation and inclusion of uncertainty quantification for risk-aware decision-making should be a priority.

5.3. Intelligent Nutrient and Irrigation Control

One of the most important operational layers in VF is intelligent nutrient and irrigation control. This is particularly true in hydroponic, aeroponic, and aquaponic systems, where the growth of the plants depends solely on the accurately controlled nutrients and not soil buffering. Rather than conventional irrigation scheduling with fixed time intervals or thresholds or PID-based control automation, AI- and ML-enabled fertigation systems provide adaptive feedback mechanisms that vary the dosing and delivery based on plant demand, environmental variations, and system dynamics [22,45,56,115,126]. Modern VF

systems are integrated with dense sensor networks for continuous monitoring of solution chemistry and hydraulic status. Most frequently measured variables are pH, electrical conductivity (EC), total dissolved solids (TDS), nutrient solution temperature, dissolved oxygen, oxidation–reduction potential, water level, and, in advanced cases, concentrations of specific ions such as nitrate, phosphate, calcium, and magnesium. ML models ingest these multi-parameter data streams and learn the relationships between nutrient profiles, plant growth responses, and environmental load conditions [162]. Reported implementations in VF literature are DNN, RF, k-KNN, SVM, Bayesian networks, LSTM networks, and XGBoost for nutrient state prediction and actuator decision support. For example, VGG16 achieved 93.82% accuracy for N-P-K deficiency detection in basil hydroponics [163,164], DNN achieved 88.5% and 81% for control action prediction in tomato and lettuce NFT systems [149], and k-NN achieved 93.3% for TDS/pH/nutrient temperature prediction in lettuce NFT [165,166]. AI-assisted fertigation systems have moved beyond static EC/pH target windows to estimate nutrient demand trajectories based on crop type, phenological stage, transpiration rates, and historical uptake patterns. ML models trained on time-series sensor data have been deployed in several hydroponic and NFT systems, either on cloud platforms or directly on edge controllers like Raspberry Pi and microcontroller gateways. These models generate real-time control outputs that are used to automatically trigger dosing pumps, solenoid valves, nutrient injectors, and recirculation units, thus creating closed-loop nutrient management architectures. Several studies indicate that ML-based fertigation control reduces over-dosing and under-supply compared to rule-based controllers. Documented benefits include lower peak energy use (20.4% water level/temperature savings and 82.1% light savings via RF), more stable EC profiles, less nutrient waste, and measurable water savings when adaptive irrigation is used in place of schedule-based irrigation. Furthermore, the nutrient use efficiency of AI-controlled systems has been improved and the risk of nutrient leakage reduced, which is of particular importance for sustainability and discharge compliance in recirculating systems. From bulk EC monitoring, more advanced approaches have evolved, such as the integration of ion-selective electrodes and spectroscopic sensing with ML regression models for ion-specific concentration estimation. Hybrid AI models have been used to estimate the concentrations of calcium, phosphate, and magnesium with acceptable ranges of RMSE (12.5, 12.1, and 7.5 mg/L, respectively), which would allow nutrient correction at the ion level instead of the aggregate conductivity level. This represents a substantial enhancement in precision compared to traditional monitoring methods, which cannot differentiate nutrient composition [22,129,141,166]. In summary, AI-enabled intelligent nutrient and irrigation management shifts fertigation from a reactive maintenance practice to a predictive and optimization-driven management practice, promoting yield stability while minimizing input waste and operational variability.

Critical Assessment: Some challenges exist in translating nutrient control models from the laboratory to the commercial environment. Ion-selective electrodes are susceptible to fouling and drift, necessitating frequent recalibration, which may not be practical in production environments. Existing ML models trained on limited sensor modalities cannot capture the complexity of nutrient–plant interactions involving multiple ions, root exudates, and microbial communities. In addition, most of the existing implementations are crop-specific and have not been validated for different cultivars and growth stages. Whether AI-assisted nutrient control systems are economically viable for small-scale operators remains to be seen.

5.4. Computer Vision and Automated Crop Monitoring

Computer vision constitutes one of the most rapidly expanding AI domains in VF due to the inherently image-rich and structured environment of multi-layer crop production systems. Camera networks deployed across racks, growth chambers, and conveyor lines enable continuous, non-contact crop monitoring at spatial and temporal resolutions that are not achievable through manual scouting. When combined with ML and DL models, these visual data streams enable automated crop assessment, early stress detection, disease recognition, growth tracking, and harvest readiness estimation [152]. Vision-based VF systems commonly employ RGB, multispectral, and thermal cameras to capture plant imagery under controlled illumination. DL architectures especially CNN, MobileNet variants, ResNet families, U-Net segmentation networks, YOLO detectors, and Faster R-CNN frameworks are widely reported for disease detection, leaf segmentation, abnormality identification, and canopy analysis. These models extract hierarchical visual features related to color distribution, venation patterns, lesion texture, shape deformation, and canopy density to infer plant condition. Across hydroponic and aeroponic VF studies, DL-based disease detection systems frequently report high classification performance, with CNN and DCNN variants achieving accuracies often above 95% under controlled datasets.

Lightweight architectures such as MobileNet V2 are increasingly preferred for edge deployment because they provide strong accuracy with lower computational and energy demand, enabling on-device inference near the sensing point [158,159]. Beyond disease detection, computer vision is also used for growth monitoring and phenotyping. Image-based models estimate leaf area, biomass proxies, growth rate, and morphological parameters using regression CNNs and mask-based segmentation networks such as Mask R-CNN; these systems enable non-destructive measurement and high-frequency growth tracking, supporting dynamic growth modeling and yield forecasting workflows [167,168]. Vision analytics further support automated sorting, grading, and selective harvesting decisions. Models trained on leaf color and texture features are used to separate abnormal leaves, detect nutrient deficiency symptoms, and identify sub-health regions in crops. Computer vision is also integrated into robotic harvesters, where visual detection guides manipulation and picking operations. Despite strong performance in controlled experiments, robustness remains a technical challenge. Model generalization across crop species, lighting spectra (especially under LED combinations), and growth stages requires large and diverse training datasets. Reported mitigation strategies include spectral normalization, data augmentation, transfer learning, and domain adaptation techniques [24,31,41,42,94,132,169].

Critical Assessment: While controlled lighting in VF is useful for consistently capturing images, it can also result in overfitting of models trained under specific spectral distribution. As LED technologies evolve and farms change their spectral recipe, models may need retraining. Practical issues such as occlusion due to overlapping leaves and mechanical interference from robotic systems also remain. The absence of standardized image datasets for VF (such as ImageNet for general computer vision) hinders model development and benchmarking. Publicly available datasets containing annotated disease images across multiple cultivars, lighting conditions, and growth stages would greatly help speed up progress.

5.5. AI-Enabled Climate and Environmental Control

Environmental regulation is one of the most technically demanding components of VF, because crop productivity in stacked indoor systems depends entirely on artificial microclimate stability rather than natural atmospheric buffering. Unlike conventional greenhouses that tolerate moderate environmental fluctuations, VF installations operate under tightly bounded setpoints for temperature, relative humidity, vapor pressure deficit

(VPD), CO_2 concentration, airflow velocity, and light spectrum. Even small deviations can propagate rapidly across multi-layer racks and lead to non-uniform growth, physiological stress, or disease susceptibility. For this reason, recent VF systems increasingly employ AI-enabled climate control frameworks that move beyond fixed threshold or PID-based automation toward predictive, adaptive, and model-driven regulation [54,148,170]. AI-enabled environmental control systems are built on dense IoT sensing infrastructures that stream high-frequency telemetry from distributed sensor nodes. Typical monitored variables include dry-bulb and canopy temperature, relative humidity, dew point, VPD, CO_2 concentration, air velocity, differential pressure, substrate temperature, photosynthetic photon flux density (PPFD), daily light integral (DLI), and spectral distribution indices. In advanced deployments, hyperspectral light sensors and plant-level thermal imaging are also integrated. These multi-modal time-series data streams form the input layer for ML models that learn the dynamic coupling between environmental drivers, plant physiological response, and control system actuation. ML models are used not only to estimate current environmental state but also to forecast short-term environmental trajectories under varying crop load, transpiration rates, and lighting schedules. Reported approaches include multivariate regression models, Random Forest and Gradient Boosting ensembles, Support Vector Regression, and deep sequence models such as LSTM and gated recurrent networks for temporal prediction. These models are particularly effective in VF because environmental dynamics are strongly time-dependent and influenced by cyclic lighting and irrigation patterns. By forecasting near-term drift in humidity, temperature, or CO_2 depletion, AI controllers can initiate preemptive corrective actions rather than reactive responses after threshold violation. AI-driven climate control coordinates multiple subsystems simultaneously, including HVAC units, chilled water loops, variable-speed fans, evaporative coolers, desiccant dehumidifiers, CO_2 injection systems, and motorized dampers. Instead of independent rule-based actuation, AI control layers perform multi-objective optimization, balancing crop physiological requirements with energy consumption constraints. Lighting control is another major domain of AI-assisted environmental regulation. In VF systems, LED arrays represent a dominant energy load, and light is both an environmental and metabolic control variable. ML-based lighting controllers dynamically adjust photoperiod, intensity (PPFD), and spectral composition (red/blue/green/far-red ratios) based on crop species, developmental stage, and observed growth response. Predictive light scheduling models have been reported to maintain target daily light integrals while minimizing instantaneous peak load, thereby improving energy efficiency and grid compatibility. Some systems incorporate RL agents that iteratively learn optimal lighting and climate strategies through reward functions tied to growth rate and energy cost. More advanced AI climate frameworks integrate external signals such as weather forecasts, electricity price curves, and grid carbon intensity into their optimization models. Even though VF is indoor, outside temperature and humidity influence HVAC efficiency and heat rejection performance. AI systems that incorporate these exogenous variables can shift cooling or CO_2 enrichment timing to improve system efficiency. Compared with classical PID or rule-based controllers, AI-assisted climate systems have been reported to show reduced oscillation, tighter setpoint tracking, improved spatial uniformity across racks, and measurable reductions in energy use per unit biomass produced [22]. Despite these advantages, challenges remain in model transferability and robustness. Climate models trained on one crop type or rack configuration may not generalize directly to others due to differences in canopy density and transpiration behavior. Sensor drift, calibration error, and communication latency can also degrade model performance. Consequently, hybrid architectures that combine physics-based control with ML prediction layers are increasingly proposed as more robust alternatives [33,37,56,60,152,171–173].

Critical Assessment: Closed-loop AI climate control in action. A network of 20 temperature/humidity sensors streams data every 5 s to an LSTM model that predicts a 2 °C rise within 3 min, triggering preemptive fan speed increases and chiller setpoint adjustments to maintain VPD within +/−0.05 kPa. The reliability of such systems, however, critically depends on the calibration of sensors and the integrity of communications. In practice, a sensor drift of ±0.5 °C can have a large impact on VPD calculations and control decisions. The energy savings found in academic studies (often 10–30%) may not directly apply to commercial operations with different grid mixes, crop cycles, and maintenance schedules. Validation needs to be performed over multiple seasons and crop rotations.

5.6. Robotics and AI-Driven Physical Automation

Robotics integrated with AI represents the extension of VF intelligence from digital monitoring and control into physical task execution. While early VF automation focused primarily on sensing and environmental regulation, recent developments show increasing deployment of AI-guided robotic systems for operational tasks such as seeding, transplanting, phenotyping, pruning, harvesting, sorting, cleaning, and infrastructure maintenance. This shift is driven by the high labor intensity of multi-layer indoor farms and the need for repeatable, high-precision manipulation in confined rack geometries. AI-enabled VF robotics relies fundamentally on perception–decision–actuation loops powered by computer vision and ML models. Vision systems mounted on robotic arms, gantry systems, or mobile platforms capture high-resolution plant imagery, which is processed using CNN, object detection (YOLO, Faster R-CNN), and segmentation networks to localize plant organs, estimate maturity, and detect abnormalities. These perception outputs feed motion planning and manipulation modules that determine grasp points, cutting locations, and harvesting trajectories. Compared with fixed automation, AI-guided perception enables selective and plant-specific intervention, which is critical for crops with non-uniform growth patterns [33,37,54,56,60,148,152,171–173]. Harvesting robots are among the most studied VF robotic applications. AI-guided harvesters identify ripeness using color, morphology, and size features and execute selective picking to avoid damage to adjacent plants. Studies report that combining deep vision models with adaptive grasp control significantly reduces mechanical injury compared with blind mechanical harvesting. Some systems further integrate yield estimation models to prioritize harvest order and optimize robotic routing paths, thereby reducing travel distance and energy use. Robotic platforms are also increasingly used for automated crop inspection and phenotyping. Rail-mounted or wheeled robots equipped with RGB, multispectral, and thermal sensors traverse VF aisles to collect standardized plant data. Integrated ML models perform on-board disease screening, growth measurement, and stress detection, enabling continuous, high-frequency crop assessment without human scouting. This approach supports early intervention and reduces the risk of undetected outbreaks spreading across layers. Maintenance automation is another emerging area. AI-driven robots are being developed for tray handling, surface cleaning, UV disinfection, and infrastructure inspection. By combining anomaly detection models with robotic inspection, systems can identify leaks, clogged emitters, or lighting failures and initiate corrective workflows. Some prototypes integrate predictive maintenance ML models that forecast component failure probability based on vibration, current draw, and operating history. From an operational perspective, robotics and AI-driven physical automation offer several advantages, including labor cost reduction, task repeatability, 24 h operation capability, and reduced human contamination risk in controlled environments. However, reviewed studies consistently highlight practical constraints that currently limit full-scale adoption. These include high capital investment, integration complexity with existing rack layouts, manipulation difficulty in dense foliage, sensor occlusion, and safety

requirements for human–robot collaboration. In addition, many robotic solutions remain crop-specific, reducing cross-crop scalability. As a result, hybrid human–robot operational models are presently more common than fully autonomous VF facilities [22,54,75,148].

Critical Assessment The case for robotic automation in VF is not yet clear-cut. Labor savings are frequently advertised; however, the capital cost of robotic systems (often hundreds of thousands to millions of dollars) requires significant throughput to achieve payback. Furthermore, the manipulation of sensitive crops like berries or leafy greens in dense configurations is technically difficult. Vision-based grasping is unreliable due to occlusion from overlapping leaves and variability in plant morphology. There are other barriers to implementation as integration of robotics into existing farm management systems and workflows is required. Until these technical and economic challenges are overcome, fully autonomous VF facilities will probably be limited to high-value crops in well-capitalized operations.

5.7. Overall Impact of AI and ML in Vertical Farming

The use of AI and ML in VF has demonstrated significant potential to enhance operational accuracy, resource efficiency, and risk management across the surveyed literature. AI-enabled VF systems constitute closed-loop digital cultivation environments with tightly coupled sensing, prediction, and actuation. Documented benefits include earlier disease detection, more accurate yield forecasting, adaptive nutrient control, optimized climate regulation, and reduced labor dependence. At the same time, persistent limitations are reported, such as the fragmentation of datasets, interoperability challenges across sensor vendors, cybersecurity risks in connected farm infrastructures, model transferability across crops, and high initial investment costs. Standardized data frameworks, cross-crop model validation, and scalable edge AI deployment strategies will be key for future progress [33,174].

6. Case Studies: AI-Driven Commercial Vertical Farming Enterprises

Over the past decade, VF has quietly moved from glossy concept art and pilot projects into real-world commercial operations. What made that leap possible, more than anything else, has been the convergence of AI, machine learning, cheap sensors, and robotics. But not every company took the same path. The five enterprises described below and a sixth that does things quite differently show how varied the business models, crop choices, and technical stacks can be. I have drawn on company reports, peer-reviewed papers, and industry updates from 2023 to 2024, not just to celebrate the successes but also to hint at the very real economic stresses that still plague this sector. Representative commercial AI-enabled vertical farming enterprises, including Aero-Farms, Infarm, Gotham Greens, Oishii, Plenty, and iFarm, illustrating diverse architectural models such as aeroponic warehouses, modular in-store farms, controlled-environment greenhouses, ro-botics-assisted fruit production systems, and AI-driven large-scale indoor farms were shown in Figure 6.

6.1. AeroFarms—Data-Driven Aeroponic Vertical Farming (USA)

One of the first commercial scale aeroponic VF operators, AeroFarms was founded in Newark, New Jersey in 2004. The company uses a proprietary aeroponic system, combined with thousands of sensors and ML models that create crop-specific recipes for growth [175]. Their flagship Newark facility opened in 2016 and is 69,000 sq ft, billed as the world's largest VF, followed by an R&D-focused site in Danville, Virginia and a 95,000 sq ft production farm in Compton, California [22,54]. AeroFarms produces over 300 varieties of leafy greens, microgreens, herbs, and salad mixes, with a yield 390 times greater per square foot than traditional field agriculture, in as little as 16–20 days [75]. The sustainability claims include 95% water reduction, zero pesticide use and significantly reduced land

footprint [35,63,172,176]. Location near major urban markets means less transport, less food miles and less spoilage. AeroFarms has been a Certified B Corporation for many years, a testament to its commitment to environmental and social governance standards.



Figure 6. Representative commercial AI-enabled vertical farming enterprises, including AeroFarms, Infarm, Gotham Greens, Oishii, Plenty, and iFarm, illustrating diverse architectural models such as aeroponic warehouses, modular in-store farms, controlled-environment greenhouses, robotics-assisted fruit production systems, and AI-driven large-scale indoor farms.

Critical Assessment: AeroFarms showed its technical sophistication and operational capability but filed for Chapter 11 bankruptcy in 2023 and restructured in 2024, exposing the financial fragility of the pure-play VF model despite technological leadership [175,176]. Aeroponic systems are still a large contributor to operational costs in terms of energy intensity, especially LED lighting and high-pressure misting. Independent verification of yield and resource efficiency claims is limited, and the bulk of data comes from company reports, not third-party audits. The long-term economic viability of the aeroponic model at scale is not yet proven.

6.2. Infarm—Distributed Modular Vertical Farming Networks (Germany/Global)

Founded in Berlin in 2013, Infarm took a distributed modular approach, deploying small, fully automated hydroponic growing units inside supermarkets, restaurants, and distribution centers [22,54]. This distributed model is particularly well suited to expensive European cities with limited land and a strong consumer interest in locally grown produce. Each unit is hydroponic, cloud-connected, and equipped with sensors to monitor light, temperature, humidity and nutrients, and an AI platform ("farm brain") for real-time adjustments [35]. Infarm has over 1400 in-store modules in ten countries, plus larger growing centers covering more than 100,000 sq ft. It grows over 75 different crops, including herbs, leafy greens, strawberries, tomatoes, peas and mushrooms. Reported resource

efficiency includes 95% water reduction, 99% land reduction, zero pesticides and up to 90% lower transport emissions [35,63,75,172,176]. In 2021, Infarm became the first European VF company to reach unicorn status with a valuation of over USD 1 billion and has partnered with many major retailers. The distributed model allows food to be grown and sold at the same location, virtually eliminating food miles [177,178].

Critical Assessment After the rapid scale-up and success of Infarm, a major financial contraction occurred in 2023–2024 with layoffs and a dramatic reduction in valuation [177,178]. The distributed model is appealing as it reduces food miles and involves consumers, but it carries significant logistics costs for module maintenance, supply chain coordination and quality control across geographically dispersed units. Unit economics are difficult, especially in competitive retail environments where pricing for produce is constrained. The scalability of the distributed model and the long-term profitability are open questions.

6.3. Gotham Greens—Controlled Environment Greenhouse–Vertical Hybrids (USA)

Founded in Brooklyn in 2009, Gotham Greens uses a hybrid model of recirculating hydroponics (mostly NFT) in high-tech greenhouses with climate control and supplemental LED lighting [177,178]. The company opened the first commercial-scale rooftop hydroponic farm in NYC (Greenpoint, 15,000 sq ft) and has now expanded to a network of large greenhouses in multiple states. As of 2024, Gotham Greens has 13 greenhouses with about 2 million sq ft of growing space, with facilities ranging in size from 200,000 to 540,000 sq ft in Virginia, Texas and Georgia. Key crops are leafy greens, herbs and salad kits, with recent expansion into value-added products such as dressings and dips. Reportedly the yield per acre is 20–30 times higher than produce grown in the field. Water reduction is 95%, land reduction is 97%, pesticides are zero, and renewable energy is used in most facilities [179]. Proximity to large cities means shorter supply chains that lower emissions and fresher products.

Critical Assessment: The hybrid greenhouse–VF approach uses natural sunlight and therefore has about 30–40% lower energy costs compared to pure indoor vertical farms that rely completely on artificial lighting. This hybrid model probably offers more resilience and lower operational risk. But geographic expansion is limited by regional insolation patterns, and the model is still reliant on access to appropriate greenhouse locations near urban markets. The hybrid approach may not be as technologically ambitious as pure indoor farms, but it may offer a more sustainable economic pathway.

6.4. Oishii—AI-Assisted Premium Fruit Vertical Farming (USA)

Oishii, established in 2016 and based in Jersey City, New Jersey, made a strategic distinction by focusing on premium fruits, especially high-value Japanese strawberries and tomatoes, rather than the commodity leafy-green market [33,179]. The company's growing system uses mobile vertical racks to move plants through different climate zones to reproduce ideal diurnal cycles, combined with AI, robotics, computer vision and live bee pollination for natural reproduction. The largest manufacturing plant, Amatelas Farm in Phillipsburg, New Jersey (237,500 sq ft) is mainly powered by an on-site solar array [33,179]. Signature products include Omakase strawberries (a luxury Japanese variety) and Rubi tomatoes. Reported productivity is >170× that of open fields per square foot, yielding millions of pounds per year [32,180]. Water use is reduced by about 90% and no use of pesticides is required. The company said it cut its harvest labor force by about 50% and its revenue grew tenfold from 2021 to 2024. Oishii is considered one of the world's most technologically advanced vertical fruit farms.

Critical Assessment: Oishii's premium price (USD 15–30 per pack) allows for higher margins and justifies significant capital investment in AI and robotics. But scalability to mass-market pricing has yet to be proven, and the long-term viability of premium price points as production scales remains unclear. The use of living pollinators in a controlled environment adds an additional level of biological complexity and operational risk. The economic model relies on consumers staying willing to pay premium prices for fruit grown indoors.

6.5. Plenty—Large-Scale AI and Robotics Vertical Farming (USA)

Plenty, founded in San Francisco in 2014, is among the most heavily funded VF companies globally, having raised some USD 1 billion and achieving a peak valuation of over USD 1.4 billion [32,180].

The company's technology platform includes tall vertical towers (up to 30 feet high) densely packed with plants, full-spectrum LEDs, robotics for harvesting, and extensive sensor networks feeding AI/ML models for environmental optimization. Plenty began with leafy greens but shifted to berries, especially strawberries, in a partnership with Driscoll's, the world's biggest berry marketer [180]. Key facilities are a former leafy greens farm in Compton, California, that was repurposed; Richmond, Virginia, the company's first vertical strawberry farm; Laramie, Wyoming, an R&D center; and an international site in Abu Dhabi. The company grows leafy greens, strawberries and tomatoes and has tested more than 50 crops. Reported yields are between 150 and 350× per acre over conventional farming, with 90–95% water reduction and zero pesticide use [35,181].

Critical Assessment Plenty's decision to close the Compton facility and refocus on berries was considered by industry observers to be a setback, demonstrating that leafy-green economics remain difficult even for well-capitalized operations [35,181]. Energy use per kg of produce is still above field production and long-term profitably will depend on premium pricing for berries and continued cost reduction in LED and automation technologies. The company's overseas expansion adds to the operational and logistical complexity. The model is technically ambitious, but its economic sustainability at scale remains to be demonstrated.

6.6. iFarm—AI Software and Turnkey Vertical Farm Platforms (Finland/International)

iFarm, launched in 2017 in Helsinki, takes a fundamentally different approach from most VF companies. Instead of operating large farms themselves, they develop and sell complete technology packages including hardware, software, and growing systems; so other businesses or farmers can set up profitable indoor farms anywhere. This "picks and shovels" strategy has been explicitly noted by investors, who see it as a more capital-efficient and faster-scaling model than building and running farms directly [182]. Their main product is Growtune, a cloud-based control platform that uses AI, machine learning, and computer vision to continuously monitor and adjust growing conditions. Using machine learning and computer vision, Growtune can detect a plant's weight as well as any growth deviations or pathologies [183]. iFarm has also developed StackGrow, a very dense vertical system with automated drones that move around the farm taking measurements. In fact, the company has equipped drones with computer vision to track crop growth and provide further data for the system's algorithm [182,184].

iFarm has pilot and demonstration farms in Switzerland, Qatar, and a few other locations. For instance, the company announced the start of construction on a robotic vertical farm in Switzerland in 2024. Based on the iFarm StackGrow technology, this farm will have a 1031 m^2 cultivation area and will produce several tons of ultra-fresh greens per month for major retail chains [185,186]. Compared to semi-automated vertical farms, this technology

increases plant density by 67% and doubles monthly yield and revenue (Vertical Farm Daily, 2024). The company claims energy savings of 20–50% compared to older vertical farm designs, 95% less water than conventional farming, and no pesticides [64,187]. Crops include leafy greens, microgreens, herbs, and increasingly berries and other fruits/vegetables. Setup costs are often quoted at roughly USD 1000 per square meter, with a strong emphasis on ROI when farms are managed efficiently. iFarm's business model is essentially about enabling other operators to succeed rather than building giant company-owned facilities. It is a lower-risk, asset-light strategy, but it also means their success depends on their customers' success which, given the industry's financial turbulence, is not a trivial bet.

Critical view: iFarm's model as a technology provider reduces direct financial risk compared with rivals that operate farms. But the company's success is tied to the profitability of its customers, which is not guaranteed given the financial volatility of the industry. The asset-light strategy enables quicker scaling but also exposes iFarm to reputational risk in case of customer operation failures. Performance claims should be independently verified in a variety of deployment scenarios. The model is a lower risk approach to VF commercialization, but is reliant on the success of a fragmented operator base.

Synthesis and Comparative Insights

The six enterprises studied reveal several recurring patterns and critical observations:

- Water and land savings are reported consistently All six companies report 90–95% water reductions and substantial land use efficiency gains (20× to 390× per unit area). These metrics seem to make sense in light of the physical principles involved in hydroponic/aeroponic cultivation and vertical stacking. However, independent verification is scarce.
- Energy intensity is the main economic barrier: LED lighting, HVAC and pumping systems account for a large share of operating expenses. Companies using natural sunlight (Gotham Greens) or onsite renewable generation (Oishii) have lower operating costs, but pure-play artificial light farms (AeroFarms, Plenty, Infarm) have tougher economics.
- Lack of proven economic viability at scale: No enterprise has yet demonstrated consistent, unsubsidized profitability across multiple crop types in competitive commodity markets. The financial fragility of the sector is demonstrated by the bankruptcy and restructuring of AeroFarms and the valuation contraction at Infarm [175–178].
- Differences in risk profiles by business models: Hybrid greenhouse model (Gotham Greens) and technology provider model (iFarm) are less risky than pure-play, fully artificial light vertical farms. Premium fruit specialization (Oishii) has higher margins but limited market size.
- Reported performance data needs independent validation: Most claims about resource efficiency and yield are from the companies themselves, and are not audited by third parties. Transparent, standardized reporting frameworks would allow for meaningful comparisons across enterprises and due diligence by investors
- Still, scalability limitations remain. While technical potential has been demonstrated at the individual facility level, scaling to multiple sites introduces operational complexity, supply chain coordination challenges, and quality control variability, all of which impact unit economics [35,181].

7. Advantages of Vertical Farming

7.1. The Need for Alternative Food Production Systems

Food security is a fundamental social concern that relates to people's regular access to sufficient, safe and nutritious food. In the absence of this fundamental assurance,

the consequences are multi-dimensional—higher levels of malnutrition, worse long-term health outcomes, slower economic growth and, in extreme cases, social unrest. And these pressures are increasing. Urban expansion, topsoil erosion and degradation through climate are causing prime farmland loss at an alarming rate [19,66,188]. Experts have warned for some time that agricultural systems have been under pressure to produce more from soils that are increasingly degraded and are approaching their limits.

VF provides a radically different way of producing food. Instead of competing for scarce arable land, VF moves production indoors and upwards into warehouses, shipping containers, retrofitted factories or purpose-built structures. Key environmental factors such as light spectrum, light intensity, temperature, humidity, CO_2 levels and nutrient delivery can be finely tuned, and the systems are remarkably resilient to the storms, heatwaves and erratic seasons that are increasingly disrupting outdoor agriculture. They can be a source of stable, predictable volumes all year round, with yields per square meter far exceeding those of open fields [189–191]. Vertical farms also create jobs for technicians, data analysts, maintenance crews, and sales personnel, in addition to productivity metrics. They help bolster local economies in regions where traditional agricultural employment has been declining.

7.2. Limitations of Traditional Agricultural Systems

For thousands of years, traditional farming has sustained civilizations, but the industrial models that dominate today are slowly eroding their own support system. Cutting down forests to create farmland destroys habitats and releases vast stores of carbon. Intensive tillage, chemical fertilizers and pesticides cause topsoil erosion, waterway pollution from nutrient runoff and greenhouse gas emissions (particularly nitrous oxide) from fossil fuel energy embedded in synthetic inputs [192–194]. In many areas, yields are either stagnant or declining, due to soil depletion and the increasing cost of inputs. Farmers are coping with more erratic weather, volatile markets and the constant pressure to produce more on less land and water. Conventional systems will face many challenges to meet the needs of a growing global population under climate change without fundamental change [188,195,196].

7.3. Increased Productivity and Consistent Yields

A very practical reason for VF is the very high degree of environmental control that it provides to growers. Light spectrum and intensity, temperature, humidity, CO_2 levels and nutrient mix can be finely tuned to optimize plant growth. This precision results in quicker growth cycles, healthier plants, and yields per unit area that are significantly higher than in open-field agriculture [160,197]. You can grow multiple crops a year with indoor farming, regardless of the season and the weather. Leafy greens that take 45–60 days to grow outdoors can be ready in 25–35 days in a well-managed vertical farm. The result is not only an increase in output per unit area, but also an increase in consistency in both quality and quantity, traits which are highly desired by retailers, restaurants and food processors [8,10,124,198]. These gains in efficiency are further amplified when combined with robotics, continuous AI monitoring and automated nutrient dosing.

7.4. Resilience and Adaptation to Climate Change

According to the World Meteorological Organization, 2023 was 1.54 °C warmer than pre-industrial times and extreme weather events are happening more often and with more ferocity. Floods, droughts, heatwaves and unseasonal storms regularly destroy open-field harvests [198]. VFs, on the other hand, are not at the mercy of external weather. Inside, plants continue to grow in stable, optimized conditions, whether it is torrential rain, extreme heat or freezing temperatures. This isolation also greatly reduces pest and

disease pressure, allowing growers to reduce or eliminate chemical controls [29,75,199]. Many vertical systems are modular, so that the capacity can be scaled-up when demand is changing or when there are regional disruptions. VF, along with drastically reduced water and land needs, represents a practical, scalable way to adapt food production to a warming, less predictable climate [39,200].

7.5. Urban Pressure Relief and the Localization of Food Systems

More than half the world's population (56.9% in 2022, over 80% in many developed countries) now lives in urban areas and the proportion is continuing to increase. As cities grow, land is taken from agriculture to provide housing and roads. At the same time, urban consumers want fresher, more sustainably grown food, with fewer food miles. This is a suitable context for VF that employs rooftops, empty warehouses, old factories, or purpose-built structures that would otherwise remain unused [16,47,201]. Local food production reduces transport emissions, decreases the risk of spoilage during long-distance transport, and delivers produce to consumers in hours instead of days or weeks [202,203]. Consumers in many cities are willing to pay a premium for food that is known to have been grown locally and sustainably. Vertical farms hold the promise of changing urban "food deserts" into areas with reliable access to fresh produce and have become tangible elements of urban revitalization and community development projects [27,66].

7.6. Benefits for Public Health

Traditional agricultural practices pose a threat to human health due to the heavy use of chemical fertilizers and untreated manure, which can contaminate drinking water and food chains with nitrates, pathogens and heavy metals [39,165]. Pesticide runoff washes into waterways, and downriver communities are exposed to long-term risks. Vertical farms are closed-loop systems with almost no runoff. The pest pressure is usually less in a controlled indoor environment which enables the growers to use less or no chemical sprays [191,194,204]. Also, with the tight control, it is also easier to focus on nutrient density rather than bulk yield, which can provide consumers with more nutritionally rich produce. In a period in which urban malnutrition and diet related diseases still represent important challenges, this nutritional quality is of great importance.

7.7. Lower Impact on Environment and Ecosystem Services

Ecosystem degradation, including forest clearance, habitat fragmentation, wetland drainage and pollution of waterways, is a major driver of modern industrial agriculture [204]. For example, millions of km^2 in the Amazon have been lost to cattle ranching and soy cultivation [29,199]. Vertical farming is an alternative paradigm. VF can release land for reforestation, biodiversity restoration, or carbon sequestration through indoor and vertical farming of food [203]. And with no field runoff, nutrient pollution is minimized. Most systems recycle nearly all water and nutrients. In urban areas, vertical farms may help to alleviate the urban heat island effect, through green infrastructure and reducing long-haul trucking [65].

8. Disadvantages of Vertical Farming (And Realistic Ways to Overcome Them)

It would be disingenuous to present VF as a panacea. The technology has genuine limitations, and acknowledging them is essential for realistic assessment. This section outlines the major challenges and the strategies being developed to address them.

8.1. Very High Capital Cost Upfront

It costs a lot of money to get a modern vertical farm up and running. Facilities need specialized structures, LED lighting rigs, climate control infrastructure, hydroponic or

aeroponic systems, extensive sensor networks, automation hardware and often proprietary software platforms. A major barrier to entry is the cost of a mid-sized commercial operation that can be tens of millions of dollars before the first harvest [29].

Mitigation Strategies: As the industry matures, designs become more standardized and component costs are coming down. Models of "farming-as-a-service" where operators lease, rather than purchase, equipment are lowering barriers to entry. Governments in Singapore, Japan, UAE and parts of Europe have introduced subsidies, grants and low-interest loans for VF specifically, recognizing its potential contribution to national food security. But capital intensity continues to be a fundamental limitation for small-scale operators and developing economies.

8.2. High Energy Consumption, Particularly Lighting

LED lighting is far more efficient than traditional grow lights, but the energy consumption of thousands of them operating for 12 to 18 h a day adds up. Where the electricity grid is expensive or based on fossil fuels, the carbon footprint of VF can be surprisingly high, possibly even exceeding that of field agriculture on a per kilogram basis, in the case of an unfavorable energy mix [203].

Mitigation Strategies: Operators are increasingly installing on-site solar arrays, entering into renewable energy power purchase agreements, or locating facilities in areas with abundant, low-cost renewable energy. Some designs, like Sky Greens' rotating towers, optimize exposure to natural daylight, thus reducing the need for artificial lighting [19]. Additionally, documented energy savings of 20–40% have been achieved with commercial applications of AI-based light optimization systems. However, the extent of these savings to offset the inherent energy intensity of indoor growing is context-dependent and requires careful crop-specific life cycle assessment.

8.3. Energy and Life Cycle Assessment Considerations

An important limitation that needs to be explicitly examined is the energy–water–land trade-off posed by VF. VF has a much lower land and water footprint compared to conventional agriculture, but its energy footprint—particularly for LED lighting, HVAC and pumping can be considerable. Preliminary life cycle assessments indicate energy consumption of 5–10 kWh per kg of lettuce grown in VF systems, compared to 0.2–0.5 kWh per kg for lettuce grown in the field [198]. However, the impacts on land use are reversed: VF requires about 0.02 m^2 per kilogram compared to 0.5 m^2 per kilogram for field agriculture. AI-optimized climate control and adaptive lighting schedules can help close this energy gap by 15–30%, but absolute energy savings should be compared to field agriculture using standard LCA metrics that include upstream manufacturing (sensors, electronics, structural materials), operational energy, and end-of-life disposal [198].

8.4. High Operational Complexity and Skilled Staff Needs

Running a vertical farm requires a bunch of interconnected systems, from water chemistry to climate control, nutrient dosing, lighting schedules, pest monitoring and data analytics. Minor errors such as a pH sensor getting out of calibration, a pump failure, or incorrect lighting can quickly ruin entire layers of crops. Most operations require staff with expertise in plant science and control systems engineering, which is rare [174,178].

Mitigation Strategies: Cloud-based control platforms (e.g., Growtune, Plenty's software, AeroFarms' recipes) automate many routine decisions. Centralized expert teams can support multiple sites through remote monitoring. Thanks to training programs and university partnerships, the number of qualified technicians is slowly increasing, and some companies are moving toward fully robotic seeding and harvesting to cut down on

human error [51,63,82,175]. Operational complexity remains a key constraint despite these developments, especially for new entrants and smaller operations.

8.5. Narrow Crop Range

Most commercial vertical farms still focus on fast-growing, high-value leafy greens, herbs and microgreens. Growing fruiting crops such as tomatoes, peppers and strawberries at scale indoors is more difficult and costly because they have longer growth cycles, require more light and have more complicated pollination needs. Currently, major staple crops (e.g., rice, wheat, and maize) are not economically viable in vertical systems and hence limit the contribution of VF to the global calorie production [179,180].

Mitigation Strategies We are moving quickly on research. Oishii and Plenty are already ramping up their production of strawberries, and companies are experimenting with tomatoes, peas, mushrooms and tiny fruit trees. This expansion is aided by genetic selection for compact, fast-fruiting varieties. The crop palette is anticipated to expand considerably in the next 5–10 years, but staple grains are unlikely to be economically competitive in the near term [198].

8.6. Risk of System Failure and Biosecurity Issues

Vertical farms are enclosed, tightly controlled environments, so a single failure, a power outage, a broken pump or the introduction of a pathogen can spread quickly and wipe out large portions of a crop. Closed systems can also create conditions conducive to certain pests or molds if hygiene protocols are not followed [188].

Mitigation Strategies: Modern facilities have multiple layers of redundancy, including backup generators, redundant pumps and separate climate zones. UV sterilization, positive air pressure and strict entry procedures reduce the risk of introducing pathogens. Now, AI-based early-warning systems can detect subtle changes in plant health or environmental parameters before problems become visible, allowing for preemptive interventions. But this concentration of production in closed systems brings about vulnerabilities that must be managed by strict operational protocols.

8.7. Uncertain Economic Viability at Scale

Some companies claim they can make money on leafy greens, but many vertical farms are still in the red, propped up by venture capital or government subsidies. High energy and labor costs, combined with competition from cheap imported field-grown produce, make the business case challenging in many markets [51,63,82,175]. The economic viability of VF is still based on premium pricing that is not sustainable for all market segments.

Mitigation Strategies: Declining technology costs. Farms in high-value markets such as Singapore, UAE, Japan, parts of the US and Europe can obtain premium prices for ultra-fresh, locally grown, pesticide-free produce. Governments are providing incentives as they see VF as a strategic asset to food security. In addition, companies are diversifying into higher margin products such as microgreens, herbs, berries and value added products such as salad kits to increase margins [51,63,82,175]. But even with these strategies, the path to consistent, unsubsidized profitability is still unclear for most operators.

8.8. Synthesis of Constraints and the Way Forward

Vertical farming is not the cure-all. It uses a lot of energy, it is technically demanding and it can only grow a limited number of crops economically. It costs a lot. However, each of these major disadvantages are being actively addressed through technological innovation, improved system design, policy support and strategic market positioning. These limitations are increasingly mitigated by AI and ML, including energy optimization, predictive maintenance, automated nutrient control and robotics. The extent to which these technolog-

ical advances can overcome fundamental economic and biophysical constraints to indoor cultivation will determine whether VF advances from a niche innovation to a meaningful component of the global food system, particularly in urban and climate-stressed regions.

9. Conclusions

From a speculative idea, vertical farming has evolved into a concrete and more realistic aspect of today's agricultural systems, particularly in cities and land-limited settings. The technology tackles several critical global issues at the same time: the slow disappearance of arable land, the increasing climate variation, the rising urbanization, the emissions from long-distance food transportation, and the environmental externalities of traditional field-based agriculture. VF supports large yield increases per unit area, near-complete independence from seasonal and climate variability, 90–95% reductions in water use, virtual elimination of synthetic pesticides in many systems, and much shorter supply chains, making it a compelling model for improving both food security and ecological sustainability in densely populated regions [19,66,188,189,191,194,204].

However, the approach is still at a relatively low level of maturity. Its immediate scalability and economic competitiveness with traditional open-field production for staple commodities is restricted by high capital expenditure, high energy requirements (especially for artificial lighting), operational complexity, reliance on skilled technical personnel, and a currently limited crop portfolio [29]. These constraints are neither trivial nor impossible to overcome. Continuous cost reductions in LEDs and renewable energy, progress in AI-based optimization and automation, modular and distributed system designs, the enabling of public policy frameworks in several countries, and rapid progress in expanding the range of viable crop types are closing the gap between demonstration-scale success and widespread commercial adoption [51,63,82,175].

9.1. Assessment of Technology Readiness

We synthesize these peer-reviewed studies and critically evaluate commercial operations to understand the maturity of AI-enabled VF applications across the following domains:

Commercially mature: AI-assisted disease detection with CNNs, climate control with regression and ensemble models, and basic yield forecasting are deployed in several commercial farms, with documented improvements in resource efficiency and early warning capability. Reported accuracies are often over 95% under controlled conditions [151–157], with limited cross-farm validation.

Demonstration stage (TRL 5–6): Robotics for harvesting and phenotyping, LSTM-based predictive control, and nutrient optimization with ion-selective sensors show promising laboratory and pilot results but face challenges in terms of integration, cost, and robustness in various commercial settings [33,37,54,56,60,148,152,171–173].

Emerging/research stage (TRL 3–4): Substantial methodological development and validation are required for federated learning for privacy-preserving multi-farm model training, explainable AI (XAI) for regulatory compliance and operator trust, few-shot learning for rapid adaptation to new crops, and AI-driven life cycle assessment optimization [22,128,133,134,142–145].

9.2. Existing AI Capabilities and Limitations for VF

The evidence reviewed indicates that AI and ML are very successful for perception and pattern recognition tasks, such as disease detection, growth monitoring, and environmental forecasting, where large, labeled datasets are available and the decision space is well-defined. However, the current capabilities of AI are limited in several important ways: (i) strong generalization across cultivars, lighting conditions, and farm configurations

without extensive retraining; (ii) operation in the face of sensor drift, data conflicts, and equipment failures in real-world operational settings; (iii) explainability of decisions to meet regulatory and operator trust requirements; (iv) optimization across the full economic–energy–environmental trade-off space without the use of integrated LCA frameworks. The most important research priorities for the next 3–5 years are these limitations.

9.3. Future Directions in Research

The present study suspects that the most interesting research over the next 3–5 years will be at the intersection of the following:

Data standardization and interoperability: The creation of open VF data schemas and secure federated learning architectures that enable collaborative model training without compromising proprietary operational data [45,105,108,115,118,122,123].

Lightweight edge-AI models: Further advancement of MobileNet variants, knowledge distillation techniques, and few-shot/zero-shot learning techniques that enable real-time inference on resource-constrained edge devices [129,136–138,140,141].

Explainable AI (XAI) for building trust with operators and achieving regulatory compliance. Integrating SHAP, LIME, and similar tools to derive actionable insights from complex deep learning models [23,129,136–138,146,147].

Dynamic optimization of lighting, climate and nutrient systems against real-time electricity pricing, grid carbon intensity, and comprehensive life cycle assessment metrics [22,54,148,170].

Cross-crop and cross-farm model generalizability: Systematic validation of AI models across multiple cultivars, lighting regimes, and farm configurations to ensure robustness and transferability [33,37,56,60,152,171–173].

9.4. The Future of Vertical Farming

VF is unlikely to be a complete replacement for conventional agriculture in the long run. Open-field systems, protected cultivation, agroecological practices, and emerging regenerative approaches will continue to provide the bulk of the world's staples in the foreseeable future. Rather, it seems that VF is well positioned to fill a niche in the strategic delivery of high-value, nutrient-dense, ultra-fresh produce to urban consumers, support food system resilience in climate-vulnerable regions, alleviate pressure on natural ecosystems through land sparing, and provide a controlled platform for rapid innovation in crop physiology, genetic selection, and resource use efficiency [29,75,199,203,204].

The fulfillment of VF's full potential will depend on the continued convergence of several factors: continued cost compression across hardware, energy, and software; demonstration of economic viability at commercial scale; expansion of the crop repertoire to include higher-value fruiting species; and sustained policy support that acknowledges indoor agriculture as a valid part of national and regional food security strategies [35,181,188]. With these developments, VF is expected to move from a niche innovation to a meaningful and durable contributor to urban and peri-urban food systems over the next one to two decades, especially in geographies where conventional agriculture is most severely limited [27,64,65,68,71–73].

In this review an effort has been made to give a balanced, critically evaluative review of the technological, agronomic, economic, and environmental aspects of VF, with special emphasis on the transformational role that artificial intelligence, machine learning, and IoT integration may play in overcoming current barriers. So far, the evidence points to a promising trajectory, but the road ahead is iterative improvement, rigorous evaluation, and pragmatic scaling, not unbridled optimism. The convergence of AI, IoT, and VF present a real and exciting pathway to climate adapted urban agriculture, but its ultimate success

is not only dependent on algorithmic accuracy but on a sustained interdisciplinary effort involving agronomy, data science, energy systems, and socio-economic planning.

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References

1. Lambin, E.F. Global Land Availability: Malthus versus Ricardo. *Glob. Food Sec.* **2012**, *1*, 83–87. [CrossRef]
2. Lal, R. Restoring Soil Quality to Mitigate Soil Degradation. *Sustainability* **2015**, *7*, 5875–5895. [CrossRef]
3. Lehman, R.M.; Cambardella, C.A.; Stott, D.E.; Acosta-Martinez, V.; Manter, D.K.; Buyer, J.S.; Maul, J.E.; Smith, J.L.; Collins, H.P.; Halvorson, J.J.; et al. Understanding and Enhancing Soil Biological Health: The Solution for Reversing Soil Degradation. *Sustainability* **2015**, *7*, 988–1027. [CrossRef]
4. Hime, N.; Vyas, A.; Lachireddy, K.; Wyett, S.; Scalley, B.; Corvalan, C. Climate Change, Health and Wellbeing: Challenges and Opportunities in NSW, Australia. *Public Health Res. Pract.* **2018**, *28*, e2841824. [CrossRef] [PubMed]
5. Arulmozhi, E.; Bhujel, A.; Moon, B.-E.; Kim, H.-T. The Application of Cameras in Precision Pig Farming: An Overview for Swine-Keeping Professionals. *Animals* **2021**, *11*, 2343. [CrossRef] [PubMed]
6. Yuan, G.N.; Marquez, G.P.B.; Deng, H.; Iu, A.; Fabella, M.; Salonga, R.B.; Ashardiono, F.; Cartagena, J.A. A Review on Urban Agriculture: Technology, Socio-Economy, and Policy. *Heliyon* **2022**, *8*, e11583. [CrossRef] [PubMed]
7. Bhat, T.A.; Mohd, S.; Mir, N.B.; Naikoo, R.H.; Kanth, F.A.; Bahar, A.; Bhat, A.; Nazir, S.; Sheraz Mahdi, Z.; Amin, L.; et al. Vertical Farming: The Future of Agriculture: A Review. *Pharma Innov. J.* **2022**, *11*, 135.
8. Food and Agriculture Organization. *The State of Food and Agriculture 2016: Climate Change, Agriculture and Food Security*; Food & Agriculture Organization of the United Nations (FAO): Rome, Italy, 2016.
9. Arulmozhi, E.; Deb, N.C.; Tamrakar, N.; Kang, D.Y.; Kang, M.Y.; Kook, J.; Basak, J.K.; Kim, H.T. From Reality to Virtuality: Revolutionizing Livestock Farming through Digital Twins. *Agriculture* **2024**, *14*, 2231. [CrossRef]
10. Food and Agriculture Organization. *Looking Ahead in World Food and Agriculture: Perspectives to 2050*; Food & Agriculture Organization of the United Nations (FAO): Rome, Italy, 2012.
11. Vermeulen, S.J.; Campbell, B.M.; Ingram, J.S.I. Climate Change and Food Systems. *Annu. Rev. Environ. Resour.* **2012**, *37*, 195–222. [CrossRef]
12. Kim, Y.; Eisenberg, D.A.; Bondank, E.N.; Chester, M.V.; Mascaro, G.; Underwood, B.S. Fail-Safe and Safe-to-Fail Adaptation: Decision-Making for Urban Flooding under Climate Change. *Clim. Change* **2017**, *145*, 397–412. [CrossRef]
13. Moghim, S.; Garna, R.K. Countries' Classification by Environmental Resilience. *J. Environ. Manag.* **2019**, *230*, 345–354. [CrossRef] [PubMed]
14. Nassar, U.A.; El-Samaty, H.S.; Waseef, A.A. Identifying Urban Factors Affecting Resilience Strategies to Withstand Sea-Level Rise in Coastal Cities: Case Study of Port Said, Egypt. *Int. J. Environ. Sustain.* **2019**, *15*, 35–53. [CrossRef]
15. Bush, J.; Doyon, A. Building Urban Resilience with Nature-Based Solutions: How Can Urban Planning Contribute? *Cities* **2019**, *95*, 102483. [CrossRef]

16. Specht, K.; Zoll, F.; Schümann, H.; Bela, J.; Kachel, J.; Robischon, M. How Will We Eat and Produce in the Cities of the Future? From Edible Insects to Vertical Farming—A Study on the Perception and Acceptability of New Approaches. *Sustainability* **2019**, *11*, 4315. [CrossRef]
17. Martin, M.; Molin, E. Environmental Assessment of an Urban Vertical Hydroponic Farming System in Sweden. *Sustainability* **2019**, *11*, 4124. [CrossRef]
18. Kalantari, F.; Tahir, O.M.; Joni, R.A.; Fatemi, E. Opportunities and Challenges in Sustainability of Vertical Farming: A Review. *J. Landsc. Ecol.* **2018**, *11*, 35–60. [CrossRef]
19. Kalantari, F.; Mohd Tahir, O.; Mahmoudi Lahijani, A.; Kalantari, S. A Review of Vertical Farming Technology: A Guide for Implementation of Building Integrated Agriculture in Cities. *Adv. Eng. For.* **2017**, *24*, 76–91. [CrossRef]
20. Al-Chalabi, M. Vertical Farming: Skyscraper Sustainability? *Sustain. Cities Soc.* **2015**, *18*, 74–77. [CrossRef]
21. Jürkenbeck, K.; Heumann, A.; Spiller, A. Sustainability Matters: Consumer Acceptance of Different Vertical Farming Systems. *Sustainability* **2019**, *11*, 4052. [CrossRef]
22. Siregar, R.R.A.; Seminar, K.B.; Wahjuni, S.; Santosa, E. Vertical Farming Perspectives in Support of Precision Agriculture Using Artificial Intelligence: A Review. *Computers* **2022**, *11*, 135. [CrossRef]
23. Popkova, E. Vertical Farms Based on Hydroponics, Deep Learning, and AI as Smart Innovation in Agriculture. In *Smart Innovation in Agriculture*; Springer Nature: Singapore, 2022; pp. 257–262.
24. Siropyan, M.; Celikel, O.; Pinarer, O. Artificial Intelligence Driven Vertical Farming Management System. In Proceedings of the World Congress on Engineering, London, UK, 6–8 July 2022.
25. Xi, L.; Zhang, M.; Zhang, L.; Lew, T.T.S.; Lam, Y.M. Novel Materials for Urban Farming. *Adv. Mater.* **2021**, *34*, e2105009. [CrossRef] [PubMed]
26. Preethi Rajan, P.R.; Lada, R.R.; MacDonald, M.T. Advancement in Indoor Vertical Farming for Microgreen Production. *CABI Databases* **2019**, *10*, 1397–1408. [CrossRef]
27. Despommier, D.; Carter, M.; Giacomelli, G.A. *The Vertical Farm: Feeding the World in the 21st Century*; Thomas Dunne Books: New York, NY, USA, 2010.
28. Othmar, R. Apparatus for the Artificial Cultivation of Plants, Bacteria, and Similar Organism. U.S. Patent No. 3,254,447, 7 June 1966.
29. Despommier, D. Vertical Farms, Building a Viable Indoor Farming Model for Cities. *Field Actions Sci. Reports. J. Field Actions* **2019**, *20*, 68–73.
30. Despommier, D. Farming up the City: The Rise of Urban Vertical Farms. *Trends Biotechnol.* **2013**, *31*, 388–389. [CrossRef] [PubMed]
31. Shamshiri, R.; Kalantari, F.; Ting, K.C.; Thorp, K.R.; Hameed, I.A.; Weltzien, C.; Ahmad, D.; Shad, Z.M. Advances in Greenhouse Automation and Controlled Environment Agriculture: A Transition to Plant Factories and Urban Agriculture. *Int. J. Agric. Biol. Eng.* **2018**, *11*, 1–22. [CrossRef]
32. Kabir, M.S.N.; Reza, M.N.; Chowdhury, M.; Ali, M.; Samsuzzaman; Ali, M.R.; Lee, K.Y.; Chung, S.-O. Technological Trends and Engineering Issues on Vertical Farms: A Review. *Horticulturae* **2023**, *9*, 1229. [CrossRef]
33. Erekath, S.; Seidlitz, H.; Schreiner, M.; Dreyer, C. Food for Future: Exploring Cutting-Edge Technology and Practices in Vertical Farm. *Sustain. Cities Soc.* **2024**, *106*, 105357. [CrossRef]
34. Levenston, M. South Korean City of Suwon Has a Vertical Farm! Available online: https://cityfarmer.info/south-korean-city-of-suwon-has-a-vertical-farm/ (accessed on 20 April 2026).
35. Manning, L. Brief: Infarm Launches Modular Growing & Distribution Center to Expand Footprint Faster. Available online: https://agfundernews.com/infarm-launches-modular-growing-distribution-center-to-expand-footprint-faster (accessed on 5 April 2026).
36. Fricke, A.; Psarianos, M.; Sabban, J.; Fitzner, M.; Reipsch, R.; Schlüter, O.K.; Dreyer, C.; Vogt, J.H.-M.; Schreiner, M.; Baldermann, S. Composite Materials for Innovative Urban Farming of Alternative Food Sources (Macroalgae and Crickets). *Front. Sustain. Food Syst.* **2022**, *6*, 1001769. [CrossRef]
37. Ragaveena, S.; Shirly Edward, A.; Surendran, U. Smart Controlled Environment Agriculture Methods: A Holistic Review. *Rev. Environ. Sci. Biotechnol.* **2021**, *20*, 887–913. [CrossRef]
38. Gan, C.I.; Soukoutou, R.; Conroy, D.M. Sustainability Framing of Controlled Environment Agriculture and Consumer Perceptions: A Review. *Sustainability* **2022**, *15*, 304. [CrossRef]
39. Al-Kodmany, K. *The Vertical City: A Sustainable Development Model*; WIT Press: Southampton, UK, 2018.
40. Al-Kodmany, K. The Vertical Farm: A Review of Developments and Implications for the Vertical City. *Buildings* **2018**, *8*, 24. [CrossRef]
41. Cohen, A.R.; Chen, G.; Berger, E.M.; Warrier, S.; Lan, G.; Grubert, E.; Dellaert, F.; Chen, Y. Dynamically Controlled Environment Agriculture: Integrating Machine Learning and Mechanistic and Physiological Models for Sustainable Food Cultivation. *ACS ES&T Eng.* **2022**, *2*, 3–19.
42. Engler, N.; Krarti, M. Review of Energy Efficiency in Controlled Environment Agriculture. *Renew. Sustain. Energy Rev.* **2021**, *141*, 110786. [CrossRef]

43. Lee, G.O.; Jang, S.-N.; Kim, M.J.; Cho, D.Y.; Cho, K.M.; Lee, J.H.; Son, K.-H. Comparison of Growth Patterns and Metabolite Composition of Different Ginseng Cultivars (Yunpoong and K-1) Grown in a Vertical Farm. *Horticulturae* **2023**, *9*, 583. [CrossRef]
44. Wheeler, R.M. Improving Vertical Farming Efficiency through Dynamic Environmental Control. *Front. Sci.* **2024**, *2*, 1465403. [CrossRef]
45. Amjad, M.; Arulmozhi, E.; Shin, Y.-H.; Kang, M.-K.; Cho, W.-J. A Review of IoT and Machine Learning for Environmental Optimization in Aeroponics. *Agronomy* **2025**, *15*, 1627. [CrossRef]
46. Elanchezhian, A.; Basak, J.K.; Park, J.; Khan, F.; Okyere, F.G.; Lee, Y.; Bhujel, A.; Lee, D.; Sihalath, T.; Kim, H.T. Evaluating Different Models Used for Predicting The Indoor Microclimatic Parameters of A Greenhouse. *Appl. Ecol. Environ. Res.* **2020**, *18*, 2141. [CrossRef]
47. Specht, K.; Siebert, R.; Hartmann, I.; Freisinger, U.; Sawicka, M.; Werner, A.; Thomaier, S.; Henckel, D.; Walk, H.; Dierich, A. Urban Agriculture of the Future: An Overview of Sustainability Aspects of Food Production in and on Buildings. *Agric. Hum. Values* **2014**, *31*, 33–51.
48. Halgamuge, M.N.; Bojovschi, A.; Fisher, P.M.J.; Le, T.C.; Adeloju, S.; Murphy, S. Internet of Things and Autonomous Control for Vertical Cultivation Walls towards Smart Food Growing: A Review. *Urban For. Urban Green.* **2021**, *61*, 127094. [CrossRef]
49. Gamal, A.; Eleinen, O.A.; Eltarabily, S.; Elgheznawy, D. Enhancing Urban Resilience in Hot Humid Climates: A Conceptual Framework for Exploring the Environmental Performance of Vertical Greening Systems (VGS). *Front. Arch. Res.* **2023**, *12*, 1260–1284. [CrossRef]
50. Ghafar, A.A.; Said, I.; Fauzi, A.M.; Shariman, M.; Shai-in; Jaafar, B. Comparison of Leaf Area Index from Four Plant Species on Vertical Greenery System in Pasir Gudang, Malaysia. *Pertanika J. Sci. Technol.* **2020**, *28*, 735–748.
51. Zaid, S.M.; Perisamy, E.; Hussein, H.; Myeda, N.E.; Zainon, N. Vertical Greenery System in Urban Tropical Climate and Its Carbon Sequestration Potential: A Review. *Ecol. Indic.* **2018**, *91*, 57–70. [CrossRef]
52. Su, M.; Jie, P.; Li, P.; Yang, F.; Huang, Z.; Shi, X. A Review on the Mechanisms behind Thermal Effect of Building Vertical Greenery Systems (VGS): Methodology, Performance and Impact Factors. *Energy Build.* **2024**, *303*, 113785. [CrossRef]
53. Raneesha Madushanki, A.A.; Halgamuge, M.N.; Surangi Wirasagoda, W.A.H.; Syed, A. Adoption of the Internet of Things (IoT) in Agriculture and Smart Farming towards Urban Greening: A Review. *Int. J. Adv. Comput. Sci. Appl.* **2019**, *10*, 11–28. [CrossRef]
54. Saad, M.H.M.; Hamdan, N.M.; Sarker, M.R. State of the Art of Urban Smart Vertical Farming Automation System: Advanced Topologies, Issues and Recommendations. *Electronics* **2021**, *10*, 1422. [CrossRef]
55. Qazi, S.; Khawaja, B.A.; Farooq, Q.U. IoT-Equipped and AI-Enabled next Generation Smart Agriculture: A Critical Review, Current Challenges and Future Trends. *IEEE Access* **2022**, *10*, 21219–21235. [CrossRef]
56. Rathor, A.S.; Choudhury, S.; Sharma, A.; Nautiyal, P.; Shah, G. Empowering Vertical Farming through IoT and AI-Driven Technologies: A Comprehensive Review. *Heliyon* **2024**, *10*, e34998. [CrossRef] [PubMed]
57. Anila, M.; Daramola, O. Applications, Technologies, and Evaluation Methods in Smart Aquaponics: A Systematic Literature Review. *Artif. Intell. Rev.* **2024**, *58*, 25. [CrossRef]
58. Shi, X.; Shi, C.; Tablada, A.; Guan, X.; Cui, M.; Rong, Y.; Zhang, Q.; Xie, X. A Review of Research Progress in Vertical Farming on Façades: Design, Technology, and Benefits. *Sustainability* **2025**, *17*, 921. [CrossRef]
59. Padhiary, M.; Prasad, G.; Hoque, A.; Kumar, K.; Sahu, B. Advances in Vertical Farming: The Role of Artificial Intelligence and Automation in Sustainable Agriculture. *LatIA* **2025**, *3*, 131. [CrossRef]
60. Sowmya, C.; Anand, M.; Indu Rani, C.; Amuthaselvi, G.; Janaki, P. Recent Developments and Inventive Approaches in Vertical Farming. *Front. Sustain. Food Syst.* **2024**, *8*, 1400787. [CrossRef]
61. Aijaz, N.; Lan, H.; Raza, T.; Yaqub, M.; Iqbal, R.; Pathan, M.S. Artificial Intelligence in Agriculture: Advancing Crop Productivity and Sustainability. *J. Agric. Food Res.* **2025**, *20*, 101762. [CrossRef]
62. Rahman, M.A.; Chakraborty, N.R.; Sufiun, A.; Banshal, S.K.; Tajnin, F.R. An AIoT-Based Hydroponic System for Crop Recommendation and Nutrient Parameter Monitorization. *Smart Agric. Technol.* **2024**, *8*, 100472. [CrossRef]
63. Vîrtosu, I.; Li, C. Vertical Farming Perspective and Challenges: A Comparative Review between China and the EU. In *Proceedings of the Central and Eastern European eDem and eGov Days 2024*; ACM: New York, NY, USA, 2024; Volume 1, pp. 1–12.
64. Corvalan, C.; Hales, S.; Mcmichael, A.J. *Ecosystems and Human Well-Being: Health Synthesis*; World Health Organization: Genève, Switzerland, 2005.
65. Thomaier, S.; Specht, K.; Henckel, D.; Dierich, A.; Siebert, R.; Freisinger, U.; Sawicka, M. Farming in and on Urban Buildings: Present Practice and Specific Novelties of Zero-Acreage Farming (ZFarming). *Renew. Agric. Food Syst.* **2014**, *30*, 43–54. [CrossRef]
66. Touliatos, D.; Dodd, I.; Mcainsh, M. Vertical Farming Increases Lettuce Yield per Unit Area Compared to Conventional Horizontal Hydroponics. *Food Energy Secur.* **2016**, *5*, 184–191. [CrossRef] [PubMed]
67. Kozai, T.; Niu, G. Chapter 4—Plant Factory as a Resource-Efficient Closed Plant Production System. In *Plant Factory*; Kozai, T., Niu, G., Takagaki, M., Eds.; Academic Press: San Diego, CA, USA, 2016; pp. 69–90.
68. Waldron, D. Evolution of Vertical Farms and the Development of a Simulation Methodology. *WIT Trans. Ecol. Environ.* **2018**, *217*, 975–986. [CrossRef]

69. Chatterjee, A.; Debnath, S.; Pal, H. Implication of Urban Agriculture and Vertical Farming for Future Sustainability. *Urban Hortic. Necessity Future* **2020**, *32*, 1854–1858.
70. Maurya, P.; Jain, S.; Kumar, V.; S., A.; Kiran, B.; Singh, P.; Nishad, U.; P., S.; Kothiyal, K. Vertical Farming: An Option in Modern Food Production: A Review. *Int. J. Environ. Clim. Change* **2023**, *13*, 883–893. [CrossRef]
71. Currey, C.J. An Introduction to Pests in Hydroponic Production. Produce Grower. Available online: https://www.producegrower.com/article/an-introduction-to-pests-in-hydroponic-production/ (accessed on 7 April 2026).
72. van Delden, S.H.; SharathKumar, M.; Butturini, M.; Graamans, L.J.A.; Heuvelink, E.; Kacira, M.; Kaiser, E.; Klamer, R.S.; Klerkx, L.; Kootstra, G.; et al. Current Status and Future Challenges in Implementing and Upscaling Vertical Farming Systems. *Nat. Food* **2021**, *2*, 944–956. [CrossRef] [PubMed]
73. Sathyanarayana, S.R.; Gangadhar, W.V.; Badrinath, M.G.; Ravindra, R.M.; Shriramrao, A.U. Hydroponics: An Intensified Agriculture Practice to Improve Food Production. *Rev. Agric. Sci.* **2022**, *10*, 101–114. [CrossRef] [PubMed]
74. Eldridge, B.M.; Manzoni, L.R.; Graham, C.A.; Rodgers, B.; Farmer, J.R.; Dodd, A.N. Getting to the Roots of Aeroponic Indoor Farming. *New Phytol.* **2020**, *228*, 1183–1192. [CrossRef] [PubMed]
75. Kannan, M.; Elavarasan, G.; Balamurugan, A.; Dhanusiya, B.; Freedon, D. Hydroponic Farming—A State of Art for the Future Agriculture. *Mater. Today* **2022**, *68*, 2163–2166. [CrossRef]
76. Barbosa, G.L.; Gadelha, F.D.A.; Kublik, N.; Proctor, A.; Reichelm, L.; Weissinger, E.; Wohlleb, G.M.; Halden, R.U. Comparison of Land, Water, and Energy Requirements of Lettuce Grown Using Hydroponic vs. Conventional Agricultural Methods. *Int. J. Environ. Res. Public Health* **2015**, *12*, 6879–6891. [CrossRef] [PubMed]
77. Ellis, J. Agricultural Transparency: Reconnecting Urban Centres with Food Production. Doctoral Dissertation, Dalhousie Universitym, Halifax, NS, Canada, 2012.
78. Voss, P. Vertical Farming: An Agricultural Revolution on the Rise. *Int. J. Sci. Res. Eng. Trends* **2013**, *10*, 1934–1942.
79. Sharma, N.; Acharya, S.; Kumar, K.; Singh, N.; Chaurasia, O. Hydroponics as an Advanced Technique for Vegetable Production: An Overview. *J. Soil Water Conserv.* **2019**, *17*, 364–371.
80. Beacham, A.M.; Vickers, L.H.; Monaghan, J.M. Vertical Farming: A Summary of Approaches to Growing Skywards. *J. Hortic. Sci. Biotechnol.* **2019**, *94*, 277–283. [CrossRef]
81. Royston, R.; Maria, M.P. Vertical Farming: A Concept. *Int. J. Eng. Technol.* **2018**, *4*, 500–506.
82. Gentry, M. Local Heat, Local Food: Integrating Vertical Hydroponic Farming with District Heating in Sweden. *Energy* **2019**, *174*, 191–197. [CrossRef]
83. Martin, M.; Molin, E. Assessing the Energy and Environmental Performance of Vertical Hydroponic Farming. *IVL* **2018**. [CrossRef]
84. Stein, E.W. The Transformative Environmental Effects Large-Scale Indoor Farming May Have on Air, Water, and Soil. *Air Soil Water Res.* **2021**, *14*, 117862212199581. [CrossRef]
85. Cooper, D. GrowCube Promises to Grow Food with Ease Indoors (Hands-On). Available online: https://www.engadget.com/2013-11-08-insert-coin-growcubes-hands-on.html (accessed on 3 April 2026).
86. Martinez, M. Vertical Farming: A Bottom-up Approach. *Seattle J. Technol. Environ. Innov. Law* **2023**, *13*, 5.
87. Gopinath, P.; Vethamoni, P.I.; Gomathi, M. Aeroponics Soilless Cultivation System for Vegetable Crops. *Chem. Sci. Rev. Lett.* **2017**, *6*, 838–849. [CrossRef]
88. Otazu, V. *Manual on Quality Seed Potato Production Using Aeroponics*; International Potato Center (CIP): Lima, Peru, 2010.
89. Lakhiar, I.A.; Gao, J.; Syed, T.N.; Chandio, F.A.; Buttar, N.A. Modern Plant Cultivation Technologies in Agriculture under Controlled Environment: A Review on Aeroponics. *J. Plant Interact.* **2018**, *13*, 338–352. [CrossRef]
90. Mirzabe, A.H.; Hajiahmad, A.; Fadavi, A.; Rafiee, S. Piezoelectric Atomizer in Aeroponic Systems: A Study of Some Fluid Properties and Optimization of Operational Parameters. *Inf. Process. Agric.* **2023**, *10*, 564–580. [CrossRef]
91. Srihajong, N.; Ruamrungsri, S.; Terdtoon, P.; Kamonpet, P.; Ohyama, T. Heat Pipe as a Cooling Mechanism in an Aeroponic System. *Appl. Therm. Eng.* **2006**, *26*, 267–276. [CrossRef]
92. Krastanova, M.; Sirakov, I.; Ivanova-Kirilova, S.; Yarkov, D.; Orozova, P. Aquaponic Systems: Biological and Technological Parameters. *Biotechnol. Biotechnol. Equip.* **2022**, *36*, 305–316. [CrossRef]
93. Gooley, G.J.; Gavine, F.M. *Integrated Agri-Aquaculture Systems: A Resource Handbook for Australian Industry Development*; Rural Industries Research and Development Corporation: Wagga Wagga, Australia, 2003.
94. Lennard, W.; Goddek, S. Aquaponics: The Basics. In *Aquaponics Food Production Systems*; Goddek, S., Ed.; Springer International Publishing: Cham, Switzerland, 2019; pp. 113–143.
95. Chandramenon, P.; Aggoun, A.; Tchuenbou-Magaia, F. Smart Approaches to Aquaponics 4.0 with Focus on Water Quality—Comprehensive Review. *Comput. Electron. Agric.* **2024**, *225*, 109256. [CrossRef]
96. Ehmann, A.; Thumm, U.; Lewandowski, I. Fertilizing Potential of Separated Biogas Digestates in Annual and Perennial Biomass Production Systems. *Front. Sustain. Food Syst.* **2018**, *2*, 12. [CrossRef]
97. Marquez, G.P.B.; Takeuchi, H.; Montaño, M.N.E.; Hasegawa, T. Performance of Rice Straw as Mono- and Co-Feedstock of Ulva Spp. for Thalassic Biogas Production. *Heliyon* **2020**, *6*, e05036. [CrossRef] [PubMed]

98. Stoknes, K.; Scholwin, F.; Krzesiński, W.; Wojciechowska, E.; Jasińska, A. Efficiency of a Novel "Food to Waste to Food" System Including Anaerobic Digestion of Food Waste and Cultivation of Vegetables on Digestate in a Bubble-Insulated Greenhouse. *Waste Manag.* **2016**, *56*, 466–476. [CrossRef] [PubMed]
99. Kumar, V.; Sharma, K.V.; Kedam, N.; Patel, A.; Kate, T.R.; Rathnayake, U. A Comprehensive Review on Smart and Sustainable Agriculture Using IoT Technologies. *Smart Agric. Technol.* **2024**, *8*, 100487. [CrossRef]
100. Kamilaris, A.; Gao, F.; Prenafeta-Boldu, F.X.; Ali, M.I. Agri-IoT: A Semantic Framework for Internet of Things-Enabled Smart Farming Applications. In Proceedings of the 2016 IEEE 3rd World Forum on Internet of Things (WF-IoT), Reston, VA, USA, 12–14 December 2016; pp. 442–447.
101. Verdouw, C.; Sundmaeker, H.; Tekinerdogan, B.; Conzon, D.; Montanaro, T. Architecture Framework of IoT-Based Food and Farm Systems: A Multiple Case Study. *Comput. Electron. Agric.* **2019**, *165*, 104939. [CrossRef]
102. Agrawal, H.; Dhall, R.; Iyer, K.S.S.; Chetlapalli, V. An Improved Energy Efficient System for IoT Enabled Precision Agriculture. *J. Ambient Intell. Humaniz. Comput.* **2020**, *11*, 2337–2348. [CrossRef]
103. Zhang, X.; Zhang, J.; Li, L.; Zhang, Y.; Yang, G. Monitoring Citrus Soil Moisture and Nutrients Using an IoT Based System. *Sensors* **2017**, *17*, 447. [CrossRef] [PubMed]
104. Rodríguez-Robles, J.; Martin, Á.; Martin, S.; Ruipérez-Valiente, J.A.; Castro, M. Autonomous Sensor Network for Rural Agriculture Environments, Low Cost, and Energy Self-Charge. *Sustainability* **2020**, *12*, 5913. [CrossRef]
105. Kassim, M.R.M. IoT Applications in Smart Agriculture: Issues and Challenges. In Proceedings of the 2020 IEEE Conference on Open Systems (ICOS), Kota Kinabalu, Malaysia, 17–19 November 2020; pp. 19–24.
106. Madakam, S.; Ramaswamy, R.; Tripathi, S. Internet of Things (IoT): A Literature Review. *J. Comput. Commun.* **2015**, *3*, 164–173.
107. Aqeel-ur-Rehman; Abbasi, A.Z.; Islam, N.; Shaikh, Z.A. A Review of Wireless Sensors and Networks' Applications in Agriculture. *Comput. Stand. Interfaces* **2014**, *36*, 263–270. [CrossRef]
108. Ray, P.P. Internet of Things for Smart Agriculture: Technologies, Practices and Future Direction. *J. Ambient Intell. Smart Environ.* **2017**, *9*, 395–420. [CrossRef]
109. Morella, P.; Lambán, M.P.; Royo, J.; Sánchez, J.C. Vertical Farming Monitoring: How Does It Work and How Much Does It Cost? *Sensors* **2023**, *23*, 3502. [CrossRef] [PubMed]
110. Bhowmick, S.; Biswas, B.; Biswas, M.; Dey, A.; Roy, S.; Sarkar, S.K. Application of IoT-Enabled Smart Agriculture in Vertical Farming. In *Advances in Communication, Devices and Networking*; Bera, R., Sarkar, S.K., Singh, O.P., Saikia, H., Eds.; Springer: Singapore, 2019; pp. 521–528.
111. Castañeda-Miranda, A.; Castaño-Meneses, V.M. Internet of Things for Smart Farming and Frost Intelligent Control in Greenhouses. *Comput. Electron. Agric.* **2020**, *176*, 105614. [CrossRef]
112. Li, X.; Ma, Z.; Zheng, J.; Liu, Y.; Zhu, L.; Zhou, N. An Effective Edge-Assisted Data Collection Approach for Critical Events in the SDWSN-Based Agricultural Internet of Things. *Electronics* **2020**, *9*, 907. [CrossRef]
113. Pisanu, T.; Garau, S.; Ortu, P.; Schirru, L.; Macciò, C. Prototype of a Low-Cost Electronic Platform for Real Time Greenhouse Environment Monitoring: An Agriculture 4.0 Perspective. *Electronics* **2020**, *9*, 726. [CrossRef]
114. Arulmozhi, E.; Bhujel, A.; Deb, N.C.; Tamrakar, N.; Kang, M.Y.; Kook, J.; Kang, D.Y.; Seo, E.W.; Kim, H.T. Development and Validation of Low-Cost Indoor Air Quality Monitoring System for Swine Buildings. *Sensors* **2024**, *24*, 3468. [CrossRef] [PubMed]
115. Chehri, A.; Chaibi, H.; Saadane, R.; Hakem, N.; Wahbi, M. A Framework of Optimizing the Deployment of IoT for Precision Agriculture Industry. *Procedia Comput. Sci.* **2020**, *176*, 2414–2422. [CrossRef]
116. Sushanth, G.; Sujatha, S. IOT Based Smart Agriculture System. In Proceedings of the 2018 International Conference on Wireless Communications, Signal Processing and Networking (WiSPNET), Chennai, India, 22–24 March 2018; pp. 1–4.
117. Haris, I.; Fasching, A.; Punzenberger, L.; Grosu, R. CPS/IoT Ecosystem: Indoor Vertical Farming System. In Proceedings of the 2019 IEEE 23rd International Symposium on Consumer Technologies (ISCT), Ancona, Italy, 19–21 June 2019; pp. 47–52.
118. Belista, F.C.L.; Go, M.P.C.; Luceñara, L.L.; Policarpio, C.J.G.; Tan, X.J.M.; Baldovino, R.G. A Smart Aeroponic Tailored for IoT Vertical Agriculture Using Network Connected Modular Environmental Chambers. In Proceedings of the 2018 IEEE 10th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment and Management (HNICEM), Baguio City, Philippines, 29 November–2 December 2018; pp. 1–4.
119. Singh, S.K.; Rawat, A.K.; Deshwal, A.; Yadav, Y.; Kumar, D.; Rai, P.K.; Gangwar, V. Vertical Farming: A Modern and Sustainable Approach to Horticultural Crop Growth. *J. Sci. Res. Rep.* **2025**, *31*, 666–685. [CrossRef]
120. Codeluppi, G.; Cilfone, A.; Davoli, L.; Ferrari, G. LoRaFarM: A LoRaWAN-Based Smart Farming Modular IoT Architecture. *Sensors* **2020**, *20*, 2028. [CrossRef] [PubMed]
121. Elijah, O.; Rahman, T.A.; Orikumhi, I.; Leow, C.Y.; Hindia, M.N. An Overview of Internet of Things (IoT) and Data Analytics in Agriculture: Benefits and Challenges. *IEEE Internet Things J.* **2018**, *5*, 3758–3773. [CrossRef]
122. Glaroudis, D.; Iossifides, A.; Chatzimisios, P. Survey, Comparison and Research Challenges of IoT Application Protocols for Smart Farming. *Comput. Netw.* **2020**, *168*, 107037. [CrossRef]

123. Gupta, S.; Das, A.; Sinha, S.K. Sustainable Vertical Farming: Leveraging Machine Learning and IoT for Energy Efficiency and Productivity. In *Proceedings of the 6th International Conference on Information Management & Machine Intelligence, Jaipur, India, 23–24 December 2024*; ACM: New York, NY, USA, 2024; pp. 1–7.
124. Saranya, T.; Deisy, C.; Sridevi, S.; Anbananthen, K.S.M. A Comparative Study of Deep Learning and Internet of Things for Precision Agriculture. *Eng. Appl. Artif. Intell.* **2023**, *122*, 106034. [CrossRef]
125. Assimakopoulos, F.; Vassilakis, C.; Margaris, D.; Kotis, K.; Spiliotopoulos, D. AI and Related Technologies in the Fields of Smart Agriculture: A Review. *Information* **2025**, *16*, 100. [CrossRef]
126. Zakir, E.; Reza, M.N.; Karim, M.R.; Ulainti, G.A.P.M.; Ali, M.R.; Jin, H.; Chung, S.-O.; Kabir, M.S.N. Remote Management and Variability Assessment of Environmental Conditions for Smart Vertical Farms: A Review. *Precis. Agric. Sci. Technol.* **2025**, *7*, 114–133.
127. Chowdhury, H.; Argha, D.B.P.; Ahmed, M.A. Artificial Intelligence in Sustainable Vertical Farming. *arXiv* **2023**. [CrossRef]
128. Naser, M.Z. An Engineer's Guide to eXplainable Artificial Intelligence and Interpretable Machine Learning: Navigating Causality, Forced Goodness, and the False Perception of Inference. *Autom. Constr.* **2021**, *129*, 103821. [CrossRef]
129. Naser, M.Z.; Alavi, A.H. Error Metrics and Performance Fitness Indicators for Artificial Intelligence and Machine Learning in Engineering and Sciences. *Archit. Struct. Constr.* **2023**, *3*, 499–517. [CrossRef]
130. Talaviya, T.; Shah, D.; Patel, N.; Yagnik, H.; Shah, M. Implementation of Artificial Intelligence in Agriculture for Optimisation of Irrigation and Application of Pesticides and Herbicides. *Artif. Intell. Agric.* **2020**, *4*, 58–73. [CrossRef]
131. Hwang, Y.; Lee, S.; Kim, T.; Baik, K.; Choi, Y. Crop Growth Monitoring System in Vertical Farms Based on Region-of-Interest Prediction. *Agriculture* **2022**, *12*, 656. [CrossRef]
132. Rizkiana, A.; Nugroho, A.P.; Salma, N.M.; Afif, S.; Masithoh, R.E.; Sutiarso, L.; Okayasu, T. Plant Growth Prediction Model for Lettuce (Lactuca Sativa.) in Plant Factories Using Artificial Neural Network. *IOP Conf. Ser. Earth Environ. Sci.* **2021**, *733*, 012027. [CrossRef]
133. Vorapatratorn, S. Development of Automatic Plant Factory Control Systems with AI-Based Artificial Lighting. In *2021 13th International Conference on Information Technology and Electrical Engineering (ICITEE)*; IEEE: New York, NW, USA, 2021; pp. 69–73.
134. Story, D.; Kacira, M.; Kubota, C.; Akoglu, A.; An, L. Lettuce Calcium Deficiency Detection with Machine Vision Computed Plant Features in Controlled Environments. *Comput. Electron. Agric.* **2010**, *74*, 238–243. [CrossRef]
135. Fan, X.; Li, C. A Simulated Framework for China's Rural Revitalization Enabled by Big Data and Smart Agriculture under the Perspective of New Quality Productivity. *Discov. Artif. Intell.* **2025**, *6*, 25. [CrossRef]
136. Sun, Y.; Liu, Y.; Wang, G.; Zhang, H. Deep Learning for Plant Identification in Natural Environment. *Comput. Intell. Neurosci.* **2017**, *2017*, 7361042. [CrossRef] [PubMed]
137. Arulmozhi, E.; Basak, J.K.; Sihalath, T.; Park, J.; Kim, H.T.; Moon, B.E. Machine Learning-Based Microclimate Model for Indoor Air Temperature and Relative Humidity Prediction in a Swine Building. *Animals* **2021**, *11*, 222. [CrossRef] [PubMed]
138. Kim, S.; Heo, S. An Agricultural Digital Twin for Mandarins Demonstrates the Potential for Individualized Agriculture. *Nat. Commun.* **2024**, *15*, 1561. [CrossRef] [PubMed]
139. Botchkarev, A. A New Typology Design of Performance Metrics to Measure Errors in Machine Learning Regression Algorithms. *Interdiscip. J. Inf. Knowl. Manag.* **2019**, *14*, 045–076. [CrossRef] [PubMed]
140. Debroy, P.; Seban, L. A Tomato Fruit Biomass Prediction Model for Aquaponics System Using Machine Learning Algorithms. *IFAC-PapersOnLine* **2022**, *55*, 709–714. [CrossRef]
141. Contreras-Castillo, J.; Guerrero-Ibañez, J.A.; Santana-Mancilla, P.C.; Anido-Rifón, L. SAgric-IoT: An IoT-Based Platform and Deep Learning for Greenhouse Monitoring. *Appl. Sci.* **2023**, *13*, 1961. [CrossRef]
142. Hao, X.; Jia, J.; Gao, W.; Guo, X.; Zhang, W.; Zheng, L.; Wang, M. MFC-CNN: An Automatic Grading Scheme for Light Stress Levels of Lettuce (*Lactuca sativa* L.) Leaves. *Comput. Electron. Agric.* **2020**, *179*, 105847. [CrossRef]
143. Hegedus, C.; Frankó, A.; Varga, P.; Gindl, S.; Tauber, M. Enabling Scalable Smart Vertical Farming with IoT and Machine Learning Technologies. In Proceedings of the NOMS 2023-2023 IEEE/IFIP Network Operations and Management Symposium, Miami, FL, USA, 8–12 May 2023; pp. 1–4.
144. Alipio, M.; Dela Cruz, A.; Doria, J.D.; Fruto, R.M. On the Design of Nutrient Film Technique Hydroponics Farm for Smart Agriculture. *Eng. Agric. Environ. Food* **2019**, *12*, 315–324. [CrossRef]
145. Musa, A.; Hamada, M.; Aliyu, F.M.; Hassan, M. An Intelligent Plant Dissease Detection System for Smart Hydroponic Using Convolutional Neural Network. In *2021 IEEE 14th International Symposium on Embedded Multicore/Many-Core Systems-on-Chip (MCSoC)*; IEEE: New York, NY, USA, 2021.
146. Raju, S.; Dappuri, B.; Penmatsa, R.K.V.; Yachamaneni, M.; Verghese, D.; Mishra, M. Design and Implementation of Smart Hydroponics Farming Using IoT-Based AI Controller with Mobile Application System. *J. Nanomater.* **2022**, *2022*, 1–12. [CrossRef]
147. Jayasekara, C.; Banneka, S.; Pasindu, G.; Udawaththa, Y.; Wellalage, S.; Abeygunawardhane, P. Automated Crop Harvesting, Growth Monitoring and Disease Detection System for Vertical Farming Greenhouse. In Proceedings of the 2021 3rd International Conference on Advancements in Computing (ICAC), Colombo, Sri Lanka, 9–11 December 2021; pp. 228–233.

148. Gong, X.; Zhang, X.; Cheng, J.; Wang, H.; Wang, K.; Meng, F. I3-YOLOv8s: An Improved YOLOv8s for Infrequent Irregular Imbalanced Detection and Segmentation of Rape Stomata. *Expert Syst. Appl.* **2024**, *264*, 125759. [CrossRef]
149. Wu, Z.; Yang, R.; Gao, F.; Wang, W.; Fu, L.; Li, R. Segmentation of Abnormal Leaves of Hydroponic Lettuce Based on DeepLabV3+ for Robotic Sorting. *Comput. Electron. Agric.* **2021**, *190*, 106443. [CrossRef]
150. Kamala, K.; Alex, S. Apple Fruit Disease Detection for Hydroponic Plants Using Leading Edge Technology Machine Learning and Image Processing. In Proceedings of the 2021 2nd International Conference on Smart Electronics and Communication (ICOSEC), Trichy, India, 7–9 October 2021; Volume 10, pp. 820–825.
151. Aquino, H.; Sybingco, E.; Mendigoria, C.H.; Concepcion, R., II; Bandala, A.; Alajas, O.J.; Dadios, E.; Vicerra, R. On-Demand Healthy and Chlorotic Lactuca Sativa Leaf Classification Using Support Vector Machine in a Rotating Hydroponic System. In Proceedings of the 2022 IEEE 14th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM), Boracay Island, Philippines, 1–4 December 2022; Volume 12, pp. 1–5.
152. Mokhtar, A.; El-Ssawy, W.; He, H.; Al-Anasari, N.; Sammen, S.S.; Gyasi-Agyei, Y.; Abuarab, M. Using Machine Learning Models to Predict Hydroponically Grown Lettuce Yield. *Front. Plant Sci.* **2022**, *13*, 706042. [CrossRef] [PubMed]
153. Gang, M.-S.; Kim, H.-J.; Kim, D.-W. Estimation of Greenhouse Lettuce Growth Indices Based on a Two-Stage CNN Using RGB-D Images. *Sensors* **2022**, *22*, 5499. [CrossRef] [PubMed]
154. Reyes Yanes, A.; Martinez, P.; Ahmad, D.R. Real-Time Growth Rate and Fresh Weight Estimation for Little Gem Romaine Lettuce in Aquaponic Grow Beds. *Comput. Electron. Agric.* **2020**, *179*, 105827. [CrossRef]
155. Apat, S.; Mishra, J.; Raju, S.; Padhy, N. The Robust and Efficient Machine Learning Model for Smart Farming Decisions and Allied Intelligent Agriculture Decisions. *J. Integr. Sci. Technol.* **2022**, *10*, 139–155.
156. Åström, O.; Hedlund, H.; Sopasakis, A. Machine-Learning Approach to Non-Destructive Biomass and Relative Growth Rate Estimation in Aeroponic Cultivation. *Agriculture* **2023**, *13*, 801. [CrossRef]
157. Gul, Z.; Bora, S. Exploiting Pre-Trained Convolutional Neural Networks for the Detection of Nutrient Deficiencies in Hydroponic Basil. *Sensors* **2023**, *23*, 5407. [CrossRef] [PubMed]
158. Mehra, M.; Saxena, S.; Sankaranarayanan, S.; Tom, R.; Veeramanikandan, M. IoT Based Hydroponics System Using Deep Neural Networks. *Comput. Electron. Agric.* **2018**, *155*, 473–486. [CrossRef]
159. Nugroho, E.; Gautama Putrada, A.; Rakhmatsyah, A. Predictive Control on Lettuce NFT-Based Hydroponic IoT Using Deep Neural Network. In Proceedings of the 2021 International Symposium on Electronics and Smart Devices (ISESD), Bandung, Indonesia, 29–30 June 2021; pp. 1–6.
160. Cho, W.-J.; Kim, H.-J.; Jung, D.-H.; Kim, D.-W.; Ahn, T.I.; Son, J.-E. On-Site Ion Monitoring System for Precision Hydroponic Nutrient Management. *Comput. Electron. Agric.* **2018**, *146*, 51–58. [CrossRef]
161. Tuan, V.N.; Dinh, T.D.; Zhang, W.; Khattak, A.M.; Le, A.T.; Saeed, I.A.; Gao, W.; Wang, M. A Smart Diagnostic Tool Based on Deep Kernel Learning for On-Site Determination of Phosphate, Calcium, and Magnesium Concentration in a Hydroponic System. *RSC Adv.* **2021**, *11*, 11177–11191. [CrossRef] [PubMed]
162. Obu, U.; Sakarkar, G.; Ambekar, Y. Computer Vision for Monitor and Control of Vertical Farms Using Machine Learning Methods. In Proceedings of the 2021 International Conference on Computational Intelligence and Computing Applications (ICCICA), Nagpur, India, 26–27 November 2021; pp. 1–6.
163. Shasteen, K.C.; Kacira, M. Predictive Modeling and Computer Vision-Based Decision Support to Optimize Resource Use in Vertical Farms. *Sustainability* **2023**, *15*, 7812. [CrossRef]
164. Verma, S.; Kumar, A.; Kumari, M.; Kumar, N.S.; Hansda, S.; Saurabh, A.; Poonia, S.; Rathore, S.D. A Review on Hydroponics and Vertical Farming for Vegetable Cultivation: Innovations and Challenges. *J. Exp. Agric. Int.* **2024**, *46*, 801–821. [CrossRef]
165. Bhargava, Y.V.; Chittoor, P.K.; Bharatiraja, C.; Verma, R.; Sathiyasekar, K. Sensor Fusion Based Intelligent Hydroponic Farming and Nursing System. *IEEE Sens. J.* **2022**, *22*, 14584–14591. [CrossRef]
166. Marchant, W.; Tosunoglu, S. Robotic Implementation to Automate a Vertical Farm System. In Proceedings of the 30st Florida Conference on Recent Advances in Robotics, Boca Raton, FL, USA, 11–12 May 2107.
167. Abdurakhmonov, B.I. Vertical Farming Innovations—A Brief Overview. In *Greenhouses—Cultivation Strategies for the Future*; IntechOpen Limited: London, UK, 2025.
168. Chen, X. The Role of Modern Agricultural Technologies in Improving Agricultural Productivity and Land Use Efficiency. *Front. Plant Sci.* **2025**, *16*, 1675657. [CrossRef] [PubMed]
169. AeroFarms. Available online: https://www.aerofarms.com/about-us/ (accessed on 7 April 2026).
170. AeroFarms Indoor Vertical Farm Expansion with Qatar Free Zones and Doha Venture Capital to Scale AeroFarms Proprietary Smart AgTech to Meet Year-Round Growing Appetite in Qatar for Fresh and Locally Grown Leafy Greens. Available online: https://www.Aerofarms.Com/2022/11/17/aerofarms-and-qfza-announce-middle-east-expansion/ (accessed on 7 April 2026).
171. Infarm. Available online: https://www.infarm.io/about-us/ (accessed on 7 April 2026).

172. Greens, G. Something Unexpected Is Sprouting in Historic Chicago Neighborhood. Available online: https://www.cbsnews.com/news/gotham-greens-viraj-puri-greenhouse-farm-on-roof-of-pullman-factory-chicago/ (accessed on 7 April 2026).
173. Gotham Greens. Available online: https://www.gothamgreens.com/ (accessed on 7 April 2026).
174. Our Farms. Available online: https://oishii.com/pages/our-farms (accessed on 5 April 2026).
175. Plenty. Available online: https://www.plenty.ag/ (accessed on 7 April 2026).
176. Martyn-Hemphill, R. Vertical Farming Software Startup iFarm Raises $4m. Available online: https://agfundernews.com/vertical-farming-software-startup-ifarm-raises-4m (accessed on 5 April 2026).
177. Marston, J. IFarm Raises $4M for Its Automated Vertical Farming System. Available online: https://thespoon.tech/ifarm-raises-4m-for-its-automated-vertical-farming-system/ (accessed on 5 April 2026).
178. Dunlap, K. Farming Is Finally Ready for Robots. Available online: https://venturebeat.com/technology/farming-is-finally-ready-for-robots (accessed on 7 April 2026).
179. iFarm to Build a Robotic Vertical Farm in Switzerland. Available online: https://www.verticalfarmdaily.com/article/9646904/ifarm-to-build-a-robotic-vertical-farm-in-switzerland/ (accessed on 5 April 2026).
180. Hygrow, iFarm will Build a Robotic Vertical Farm in Switzerland. Available online: https://www.newaginternational.com/cea-and-greenhouse/ifarm-begins-construction-of-vertical-farm-in-switzerland/ (accessed on 5 July 2026).
181. Vertical Farming Technologies by iFarm. Available online: https://ifarmfi.com/technologies (accessed on 5 July 2026).
182. Lambin, E.F.; Meyfroidt, P. Global Land Use Change, Economic Globalization, and the Looming Land Scarcity. *Proc. Natl. Acad. Sci. USA* **2011**, *108*, 3465–3472. [CrossRef] [PubMed]
183. Oh, S.; Lu, C. Vertical Farming—Smart Urban Agriculture for Enhancing Resilience and Sustainability in Food Security. *J. Hortic. Sci. Biotechnol.* **2022**, *98*, 1–8. [CrossRef]
184. Rosenzweig, C.; Elliott, J.; Deryng, D.; Ruane, A.C.; Müller, C.; Arneth, A.; Boote, K.J.; Folberth, C.; Glotter, M.; Khabarov, N.; et al. Assessing Agricultural Risks of Climate Change in the 21st Century in a Global Gridded Crop Model Intercomparison. *Proc. Natl. Acad. Sci. USA* **2014**, *111*, 3268–3273. [CrossRef] [PubMed]
185. Lal, R. Soil Degradation as a Reason for Inadequate Human Nutrition. *Sci. Sociol. Econ. Food Prod. Access Food* **2009**, *1*, 45–57. [CrossRef]
186. Matson, P.A.; Parton, W.J.; Power, A.G.; Swift, M.J. Agricultural Intensification and Ecosystem Properties. *Science* **1997**, *277*, 504–509. [CrossRef] [PubMed]
187. Spielman, D.; Pandya-Lorch, R. *Millions Fed: Proven Successes in Agricultural Development*; International Food Policy Research Institute: Washington, DC, USA, 2009. [CrossRef]
188. Basarir, A.; Al Mansouri, N.M.N.; Ahmed, Z.F.R. Householders Attitude, Preferences, and Willingness to Have Home Garden at Time of Pandemics. *Horticulturae* **2022**, *8*, 56. [CrossRef]
189. Tilman, D.; Fargione, J.; Wolff, B.; D'Antonio, C.; Dobson, A.; Howarth, R.; Schindler, D.; Schlesinger, W.H.; Simberloff, D.; Swackhamer, D. Forecasting Agriculturally Driven Global Environmental Change. *Science* **2001**, *292*, 281–284. [CrossRef] [PubMed]
190. Ptak, M.; Wasieńko, S.; Makuła, P. A New Approach to Vertical Plant Cultivation Maximises Crop Efficiency. *Sustainability* **2024**, *16*, 7189. [CrossRef]
191. Zhu, X.; Marcelis, L.F.M. Vertical Farming for Crop Production. *J. Mod. Agric.* **2023**, *1*, 13–15. [CrossRef]
192. Quy, V.K.; Van Hau, N.; Van Anh, D.; Quy, N.M.; Ban, N.T.; Lanza, S.; Randazzo, G.; Muzirafuti, A. IoT-Enabled Smart Agriculture: Architecture, Applications, and Challenges. *Appl. Sci.* **2022**, *12*, 3396. [CrossRef]
193. Ceccanti, F.; Bischi, A.; Desideri, U.; Baccioli, A. Beyond Yield: Integrating Energy, Water, Cost, and Carbon to Benchmark Indoor Vertical Farming Viability. *Energy Convers. Manag.* **2026**, *348*, 120646. [CrossRef]
194. Benke, K.; Tomkins, B. Future Food-Production Systems: Vertical Farming and Controlled-Environment Agriculture. *Sustain. Sci. Pract. Policy* **2017**, *13*, 13–26. [CrossRef]
195. Wang, Y.P.; Kintrea, K. Urban Expansion and Land Use Changes in Asia and Africa. *Environ. Urban. Asia* **2021**, *12*, 097542532199908. [CrossRef]
196. Sanyé-Mengual, E.; Anguelovski, I.; Oliver-Solà, J.; Montero, J.I.; Rieradevall, J. Resolving Differing Stakeholder Perceptions of Urban Rooftop Farming in Mediterranean Cities: Promoting Food Production as a Driver for Innovative Forms of Urban Agriculture. *Agric. Hum. Values* **2016**, *33*, 101–120. [CrossRef]
197. Khalil, H.; Khalid, A.W. Advantage of Vertical Farming over Horizontal Farming in Achieving Sustainable City, Baghdad City-Commercial Street Case Study. *IOP Conf. Ser. Mater. Sci. Eng.* **2020**, *745*, 012173. [CrossRef]
198. Zaręba, A.; Krzemińska, A.; Kozik, R. Urban Vertical Farming as an Example of Nature-Based Solutions Supporting a Healthy Society Living in the Urban Environment. *Resources* **2021**, *10*, 109. [CrossRef]

199. Emmerson, M.; Morales, M.B.; Oñate, J.J.; Batáry, P.; Berendse, F.; Liira, J.; Aavik, T.; Guerrero, I.; Bommarco, R.; Eggers, S.; et al. Chapter Two-How Agricultural Intensification Affects Biodiversity and Ecosystem Services. In *Large-Scale Ecology: Model Systems to Global Perspectives*; Dumbrell, A.J., Kordas, R.L., Woodward, G., Eds.; Advances in Ecological Research; Academic Press: Cambridge, MA, USA, 2016; Volume 55, pp. 43–97.
200. Walker, B.; Steffen, W.; Bondeau, A.; Bugmann, H.; Campbell, B.M.; Canadell, P.; Cramer, W.; Ehleringer, J.; Elliott, T.; Foley, J.; et al. *The Terrestrial Biosphere and Global Change: Implications for Natural and Managed Ecosystems*; Cambridge University Press: Cambridge, UK, 1999.
201. Maheshwari, S. Vertical Farming: Resilience Towards Climate Change. In *Urban Growth and Environmental Issues in India*; Springer: Singapore, 2021; Volume 9, pp. 207–221.
202. Avgoustaki, D.D.; Xydis, G. How Energy Innovation in Indoor Vertical Farming Can Improve Food Security, Sustainability, and Food Safety? *Adv. Food Secur. Sustain.* **2020**, *5*, 1–51. [CrossRef]
203. Sky Greens-Farming in The Sky. Available online: https://singaporemagazine.sif.org.sg/Stories/Data/Stories/Farming-In-The-Sky_2015Issue3?gad_source=1&gad_campaignid=22986706454&gbraid=0AAAADnoCM7G8k6rCErLd9SGhbDRgo1Yk&gclid=Cj0KCQjws83OBhD4ARIsACblj18NSyIriX4OkRVEBXQwaE1pq_ohvZpXHXoAyoWigMFQYetTxMQvXhIaAo6vEALw_wcB (accessed on 7 April 2026).
204. EcoWatch World's Largest 'Vegetable Factory' Revolutionizes Indoor Farming. Available online: https://www.ecowatch.com/worlds-largest-vegetable-factory-revolutionizes-indoor-farming-1882004257.html (accessed on 3 April 2026).

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